Links

https://public.cyber.mil
- army pki information

https://learnbyexample.github.io/learn_gnuawk/cover.html
- % awk manual

https://eicar.org
- sample malware file for NetSec labbing

https://badssl.com
- PKI cert testing/labbing resource

https://zsh.sourceforge.io/Doc/Release/
- zShell manual

https://en.wikipedia.org/wiki/Programming_language
- programming language

https://en.wikipedia.org/wiki/Scripting_language
- list of scripting languages/extensions

https://en.wikipedia.org/wiki/List_of_file_signatures
- list of file signatures(hexdumps)

https://www.cve.org
- tracks common vulnerabilities and exploits
- labled CVE-[year]-[number]
- organized by severity using Common Vulnerability Scoring System(CVSS)

https://www.owasp.org/Top10/2025/
- tracks common web vulnerabilities

https://www.cisco.com/c/en/us/solutions/collateral/enterprise/design-zone-security/zt-ag.html
- ZTNA guide

https://www.cisco.com/c/en/us/solutions/collateral/enterprise/design-zone-security/safe-overview-guide.html
- SAFE Guide

https://github.com/CiscoDevNet/python_code_samples_network/tree/master/RESTCONF
- RESTCONF python code examples

https://www.ascii-code.com/
- ascii code understanding chart

https://www.cisco.com/c/en/us/products/collateral/security/security-cloud-control/datasheet-c78-736847.html
- Security Cloud Control Data Sheet

https://www.cisecurity.org/cis-benchmarks
- CIS Benchmarks
  **device security best practices & configs

https://www.nist.gov/publications/guide-ipsec-vpns
- tech pubs
- read understanding, follow up w/ vendor guides for implementation

https://vscode.dev
- open source scripting environment

https://github.com/AdiH8/Active-Directory-Lab

https://manpages.ubuntu.com/
- linux language

rdweb.wvd.azure.us

https://client.wvd.azure.us/arm/webclient/index.html

http://armylinks.com

https://akooffline.us

https://armypubs.army.mil/

https://lwn.army.mil/

https://github.com/GNS3/gns3-gui/releases

https://www.sysnettechsolutions.com/en/cisco-ios-download-for-gns3/

https://learn.microsoft.com/en-us/
- /docs/
- /powershell/
- /windows-server/
- /windows/

https://www.iragreen.com/

https://docs.brew.sh

overTheWire path: /tmp/hardToGuessName/

Scrum?
Agile?

Re-Up Bonus: tax free-overseas

Certifications(10pts per cert•per class): [CCNP Security][SERVER+][A+]

- education tools: compTIA - udemy

- Sgt Oliver(automations) - list of correspondence classes

MEMORIZE THESE STUDY RESOURCES

- Configuring an MPLS Network from Scratch: Kevin Wallace Training, LLC ~~~ youtube

- CCNA Topics(playlist): Kevin Wallace Training, LLC ~~~ youtube

- CCNA Routing & Switching Packet Tracer Labs ~~~ youtube

- Free CCNA v1.1 200-301 I Complete Course 2025 ~~~ youtube

- SAA ~~~ teambook

- ASA ~~~ teambook

- SYSTEMS ~~~ finder

- Multicast vs. Broadcast vs. Anycast vs. Unicast ~~~ https://www.baeldung.com/cs/multicast-vs-broadcast-anycast-unicast

- GNS3: Start here if you are new to GNS3 ~~~ https://www.youtube.com/playlist?list=PLhfrWIILOoKNFP_e5xcx5e2GDJIgk3ep6

- Resources - Cisco Reference Articles ~~~ https://www.connecteddots.online/resources/cisco-reference

- How VRFs Work in and MPLS Network ~~~ https://www.youtube.com/watch?v=_zjH0o1RlKI

- GRE Tunnel Idea ~~~ youtube(playlist•VPC Network)

- 7 Multipoint GRE mGRE Tunnel ~~~ https://www.youtube.com/watch?v=U5A6GqMyNW4

- 5 Encrypting GRE Tunnels using IPSec ~~~ https://www.youtube.com/watch?v=Ef9W8dO9Ar8

- 8 Configuring DMVPN - Phase 1 ~~~ https://www.youtube.com/watch?v=CRA3uAxHLZM&list=PLI5_9lI6uSzIHf3-1wNvzwSjuUVd4H07D&index=1&t=3s

- How TCP really works: MTU vs MSS ~~~ https://www.youtube.com/watch?v=J-gnDC6B5eE&t=987s

- basic CMD/TERMINAL commands ~~~ https://gist.github.com/carlessanagustin

armyRemoteDesktop DN: rdweb.wvd.azure.us

https://learn.atis.army.mil: learn army domain systems

https://ncolcoe.llc.army.mil: blackboard url

https://atn.army.mil: training database

https://ncolcoe.llc.army.mil: BLC blackboard

https://newserverlife.com: use to build server(then shop used)

https://savemyserver.com: use to build server(then shop used)

https://microcenter.com: use to build server(then shop used)

https://wiresoft.us: software/hypervisor licences

https://visiocafe.com: network/cable/wiring/rack diagraming

https://gns3.com/marketplace/appliances: vms

TCP/IP Port Document - iana.org/assignments/service-names-port-numbers/service-names-port-numbers.html

Hardware Tests

Stress Testing: Cinebench

Benchmarking: 3D Mark

C2: command and control(C&C)

- handlers and bots

Agents: software packages - primarily meant for networking services - installed to promote connectivity/aggregation for remote devices

Protocol Independent Multicast(PIM)

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PKI Certificates Rundown

- A ////// B
  - simple packet

- how to ensure only B reads
  - encrypt with B public key
  - only B can decrypt with private key

- how to ensure message is unaltered
  - hash message and append to message encrypted with B public key
  - decrypted message will match the hash of As plain texted trasmission

- how to ensure sender nonrepudiation(signature) - along with aforementioned security provisions
  - hash message, sign hash with A private key, append to message encrypted with B public key
  - decrypted with As public key, the resulting hash matches that of As plain text transmission

- how to ensure sender integrity(certificate)
  - register PKI with a certificate authority
  - third party ensures signer identity and credential validity/revocation

- how to ensure CA safe exchanges
  - utilizing common or CAs for A ////// B secure messaging

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FUOPS - future operations

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LTT Framework(refer to BLC LTT rubric)

- present the use case

- solve the use case

- explain the components of the solution

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- simplest use case to most complicated

- 2 practical exercises and an evaluation should suffice

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Basic IT Analogies

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- network address: streetname

- broadcast: message board/neighborhood group chat

- gateway: mail room

- multicast: social media

- anycast: proximal services(like food & goods delivery)

- unicast: mail

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Seamless Wireless LAN Swap

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- retain SSID

- retain password

- retain security protocol

- retain LAN architecture
    - dhcp scope & vlans

- retain port forwarding

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Traffic Testing Tools

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iperf.fr
    - simple client/server agents + large file transfer stress test

linux.die.net/man/1/ttcp

bwping.sourceforge.io

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Packet Capture Tools

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tcpdump.org

ngrep(github.com/jpr5/ngrep)
    - supports use of regular expressions found on regexr.com

nmap.org/ncat

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Keys

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x^n = P
    x = # of available digits
    n = key length
    P = total choices possible

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History

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% !13
    ! recalls commands from history entry [13]

% history -30
    ! displays the last 30 entries

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Variables, Arrays, Hashtables, Special Characters

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% variable="testString"

% unset variable
    ! clears stored value for $variable

% echo $variable
    ! displays empty value for $variable

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% arr=()
    ! declares an empty array "arr"

% arr=(foo bar baz)
    ! declares an array with three values
    % echo ${#arr[@]}
        ! displays the element count [#] for the array [arr]
        ! displays the value "3"
    % echo ${arr[0]}
        ! displays index 0
        ! displays the element "foo"

% printf '%s\n' foo bar baz
    ! displays elements [%s] "foo bar baz" separated by new lines [\n]

% printf '%s\n' "${arr[@]}"
        ! correct syntax for iteration over elements in array with formating

% IFS=,
% echo "${arr[*]}"
% unset IFS
    ! input field separator
    ! sets joining delimiter to [,] when iterating the elements of an array into a string
    ! unsets IFS for future usage with other delimiters
    **(IFS = ,; echo "${arr[*]}")
        ! future prove syntax with identical functionality

**CHEAT SHEET:
    @ /// <- iterates elements
    * /////// <- stringifies elements

++

% declare -A arr
    ! declares variable [arr] as an associative array
% arr[foo]=1
    ! declares value [1[of element [foo]
    % arr[bar]=2
% echo "${arr[foo]}
    ! index into the array [arr] as position [foo] to return its value

% printf '%s\n' "${arr[@]}"
    ! displays values [%s] "1 2 3" separated by new lines [\n]

% printf '%s\n' "${!arr[@]}"
    ! displays indices [%s] "foo bar baz" separated by new lines [\n]

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% …/n
    ! newline

% …/t
    ! tab(blank space)

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Unique

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% cat ./testFile | awk -F : '{printf("%s has shell %s\n", $1, $7) ; }' | sort | uniq
! outputs unique entries only(NO RELICATED ENTRIES)

% cat ./testFile | awk -F : '{printf("%s has shell %s\n", $1, $7) ; }' | sort | uniq -c
! outputs unique entries with a frequency count [-c]

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Sort

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% cat ./testFile | awk -F : '{printf("%s has shell %s\n", $1, $7) ; }' | sort
! sorts the data into alphabetical order

% awk '{FS="," ; OFS="," ; print $1,$2,$30,$24}' ./fantasyPerGame.csv | sort -t "," -k4,4gr
! identifies the delimiter[-t]
! sets the key [-k]
! sorts in numerical order [g] in reverse [r]

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Awk

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% cat ./testFile | awk '{ print $0 }
! prints lines in their entirety

% cat ./testFile | awk '{ print $1 }'
! prints first field of each line

% cat ./testFile | awk -F : '{ print $1 }'
! prints each line treating [:] as the delimiter
** default is a single space

% cat ./testFile | awk -F : '{printf("%s has shell %s\n", $1, $7) ; }'
ex. root has shell /bin/bash
! investigate this syntax ?

% cat ./testFile | awk -F : ' $1 == "testField" {printf("%s has shell %s\n", $1, $7) ; }'
ex. testField has shell /bin/bash
! investigate this syntax ?

% awk '{FS="," ; OFS="," ; print $1,$2,$30,$24}' ./fantasyPerGame.csv
! displays 4 fields of each record from the table provided
! ackowledges the field separatro [FS] and replaces it with another [OFS]

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Sed(stream editor)

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% cat ./testFile | sed 's/testField/replacement/'
! searches file for target "testField"
! replaces target with "replacement"
**changes what prints to the screen(DOES NOT MAKE FILE CHANGES)

% sed 's/testF…d/replacement/'
! [.] represents "some character" as opposed to rigid input

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Cut

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% cat ./testFile | cut -d : -f 1
! displays contents of file "testFile"
! separates string at character ":"
! begins display at field "1" which is the user name in this case
**does not require cat for function

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Help(built-ins)

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bash% help [built-in]
! displays manual info for built in commands
! be sure to switch to bash prior to issuing the command

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Translate

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% echo $PATH | tr : '\n'
! translates[tr] outputs colons to new lines['\n]
! [$] tells the terminal that the following is a variable and to read its contents into the comand line
**best used w/ scripting boundaries for safety

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Homebrew Install

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% /bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"

% echo >> /Users/dunnkymani/.zprofile

% echo 'eval "$(/opt/homebrew/bin/brew shellenv zsh)"' >> /Users/dunnkymani/.zprofile

% eval "$(/opt/homebrew/bin/brew shellenv zsh)"

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Git Basics

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% brew install git

% git config --global user.name "ohGeeKee"

% git config --global user.email "yaiu0472@gmail.com"

% git config --list

% git init
! current directory becomes git repository
**best practice is to create a new directory prior

% git status

```
% git add
	! stages repository unchecked changes

% git add .
	! stages all changes
	% git add test.txt
	! stages specific files

% git rm --cached <filename>
	! unstaged the added filename

% git commit -m "name of changes"
	! saves changes to branch

% git clone https://github.com/ohGeeKee/testRepository.git
	! grabs a copy of the target repository for local access

% git checkout -b newBranchName
	! creates and switches to new branch

% git branch
	! displays branches in repository

% git branch -d <branch name>
	! deletes the target branch from the repository LOCALLY
	! mustn't be in the target branch for deletion

% git push -d <remote name> <branch name>

% git switch main
	! returns to main branch
	% git switch newBranchName

% git remote add origin
	! targets github repository for push changes via HTTPS
	% git remote -v
		! verifies target repository
		**target repository has to exist on GitHub for push completion
% ls -al ~/.ssh
	! checks for existing ssh keys as password auth has been obsoleted

% ssh-keygen -t ed25519 -C "yaiu0472@gmail.com"
	! creates a new key pair for GitHub interactions

% eval "$(ssh-agent -s)"
	! output: Agent pid #####
	! starts ssh agent in the background on port #####

% open ~/.ssh/config
	! checks for the file where keys are automatically loaded and passphrases are stored

% nano config
	! Host github.com
	!	AddKeysToAgent yes
	!	UseKeychain yes
	!	IdentityFile ~/.ssh/id_ed25519
	! ## creates a SSH config file
	! ## automatically loads keys/passphrases into the ssh agent

% ssh-add --apple-use-keychain ~/.ssh/id_ed25519
	! appends passphrase to ssh-agent using keychain

% git push origin main
	! [main] interchangeable with any branch

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SSH

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% ssh bandit.labs.overthewire.org -p 2220
	- TCP SYN sent to site on port
	- [-p] specifies port 2220

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trailing slashes

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% cp -r /source /destination
	ex. copy the source directory itself and its contents

% cp -r /source/ /destination
	ex. copy ONLY the contents of the source
	**source contains trailing slash

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rsync

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- secure network and local file transfer over SSH

- equipped with "synching"  mechanisms, which check destination and perform delta transfers
	**delta transfer: existing files remain, thereby only copying the differences over

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% rsync -rv --dry-run notes jay@10.10.10.218:/home/jayRemote/backup
	- copies directory "notes" to directory located on remote user "jay" system at 10.10.10.218
	- [-r] recursive for directory copies
	- [-v] displays task dialogue
	- [--dry-run] simulates file transfer output without remote system execution
		**remove to execute

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grep

	- search file for term

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% grep -i file ./containsTheWordFile.txt
	- searches file for the word "file"
	- [-i] case insensitive

% grep -r file ./directory/multipleFiles
	- searches directory for the instances of the word "file"
	- [-r] searches recursively
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Hostname & scutil

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% hostname
	- prints system hostname

% hostname beastie.localdomain.org
	- set hostname to string following command
	**resets upon reboot

% scutil --set Hostname beatie.localdomain.org
```

- sets hostname to “beastie.localdomain.org” perpetually

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Display Free Disk Space

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% df -h
- displays volumes and memory statistics

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Basic system information

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% system_profiler
- displays basic system information
**pair with tab completion for more specified output

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% curl ifconfig.me
- displays public ip

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>systeminfo
- windows computer system info

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Powershell Basics

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>Get-PSDrive
- returns all drives on the computer

>Get-PSDrive -PSProvider Filesystem
- returns only drives with a filesystem

>Get-ChildItem
- lists files in the directory
>Get-ChildItem c:\test
- lists files in the directory of an absolute/relative path

>Set-Location c:\Test
- change directory

>New-Item -ItemType Directory -Path c:\ -Name Test2
- creates directory “Testw” on c: drive

>New-Item -ItemType File -Path c:\Test2 -Name testfile1.txt -Value “This is my file”
- creates file “testfile1.txt” containing the text “this is my file” in the directory Test2
>Set-Content -Path c:\Test2\testfile1.txt -Value “This is replaced value”
- changes textfile1.txt contents to “This is replaced value”

>Get-Item testfile1.txt
- returns file information

>Get-Content textfile1.txt
- return file contents
>Get-Content testfile1.txt | Out-Gridview
- opens contents in text editor
>Get-Content testfile1.txt | Out-Printer
- generates save-as prompt

>Remane-Item -Path c:\Test2 -NewName Test3
- changes directory name from “Test2” to ““Test3
**be sure to navigate away from Test2 before issuing the command as name will not change while directory is active

>Copy-Item -Path c:\test\testfile2.txt -Destination c:\Test3
- copies “testfile2.txt” to directory “Test3”

>Move-Item -Path c:\test\testfile4.txt -Destination c:\Test3
- moves file “testfile4.txt” to directory “Test3”

>Move-Item -Path c:\test\testfile2.txt -Destination c:\Test3\testfile5.txt
- moves and renames file “tesfile2.txt” to “testfile5.txt” located in directory “Test3”

>Remove-Item c:\Test3\testfile2.txt
- deletes file testfile2.txt

>Get-Command
- lists available commands
>Get-Command -CommandType cmdlet
- lists available cmdlets
>Get-Command *Item
- lists available commands containing the word “Item”

>Get-Help Move-Item
- returns manual
>Get-Help Move-Item -Example
- returns manual listing and example of the commands use

>Get-Process
- returns current processes

>Get-Service
- running services

>Get-Process Notepad
- returns information regarding the process named “Notepad”

>Stop-Process -Name Notepad
- ends process “Notepad”
>Stop-Process -Id 108576 -Confirm -Passthru
- ends process associated with ID “108576”

>Get-WindowsDriver -Online -All
- returns installed and running drivers
>Export-WindowsDriver -Online -Destination e:\Test\DriveBackup
- snapshots drivers to directory “DriveBackup”

>Start-MPScan -ScanType QuickScan -ScanPath e:\OldDocs
- malware scan of directory “OldDocs”

>test-netconnection 192.168.2.1 -port 21 -traceroute
- tests host at 192.168.2.1 for FTP connectivity
- uses ICMP echos to display hops between source and destination
**useful for troubleshooting connectivity issues(FWH/ACL intervention)

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mDNS

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% log stream --info --predicate 'process == "mDNSResponder"'
- displays log on mDNSResponder activity

++

% nano /etc/pf.conf

[inside packet filtering document]

block drop out proto udp from any to 224.0.0.251 port 5353
block drop out proto udp from any to ff:02:gb port 5353

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nettop

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% nettop -m tcp -P
- displays list of sockets(-m tcp), tcp
- per-process summary(-P)
**bandwidth information per connection

%nettop -n -m udp -t wifi -s 1
- displays list of active socket connections
- terminates DNS resolution
- limits output to udp ports (-m udp)
- limits monitoring to wifi interface (-t wifi)
- refresh rate of 1s(ie m/s bandwidth)[-s 1]

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NTP

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Linux NTP Server Config(chrony)

% yum -y install chrony

% vi /etc/chrony.conf

% systemctl enable chronyd

% systemctl start chronyd

% systemctl status chronyd

% chronyc sourcestats
- displays detailed information related to each time server

% chronyc sources

++

Linux NTP Server Config

% yum -y install ntp

% vi /etc/ntp.conf

% systemctl enable ntpd
- enable NTP daemon to startup at boot

% systemctl start ntpd

% systemctl status ntpd

% ntpq -p
- prints list of peers and their states
**syncing with word “pools” returns different IPs - still listed amongst the pool - than listen in the .conf

++

Windows NTP Server Configuration

- task manager\ services tab\ open services button\ windows time(right click)\ properties

- startup type: automatic

- registry editor\ hkey_local_machine\ system\ currentControlSet\ services\ w23time\ parameters\ type(right click)\\modify

- Value Data: NTP

- w32time\ time providers\ ntpServer\ enabled

- Value Data: 1

- w32time\ config\ announceFlags

- value data: 5

- task manager\ services tab\ open services button\ windows time(right click)

- restart

++

Windows NTP Client Configuration(CLI)

>w32tm /config /manualpeerlist:192.168.0.22,0x8/syncfromflags:MANUAL /update
- sets server 192.168.0.22 as NTP peer

> w32tm /query /status
- checks current NTP status

> w32tm /query /source
- checks current NTP source

> w32tm /query /peers
- checks current NTP peers

> w32tm /resync
- force a time synchronization

> w32tm /resync /rediscover
- force rediscovery of domain time sources

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public NTP servers:
- time.google.com
- time.windows.com
- time.apple.com
- time.nist.gov
- pool.ntp.org

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% /etc/ntp.conf
- configuration file for ntp

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DNS Records

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whois
%whois 8.8.8.8
- returns verbose information about the owner of the public IP

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nslookup

>nslookup -type=mx 515.support.com 8.8.8.8
- query googles DNS server(8.8.8.8), for information about 515support.com's mail records

>nslookup -type=ns comptia.org 8.8.8.8
- query googles dns server for information about comptia.org's name servers

++

% dig 515support.com MX
- queries default DNS for mail exchange resource associated with 515support.com

% dig mail.515support.com
- queries default DNS for ip address associated with "mail.515support.com"

% dig @ns1.isp.example 515support.com
- queries nameserver "ns1.isp.example.com" for ip address associated with 515support.com

% dig +trace comptia.com
- displays dns hops from root to resolved IP
**includes root, TLD, authoritative servers

++

DNS caching - Linux

%yum install dnsmasq
- install dns caching software

%cd /etc

%vim /etc/dnsmasq.conf
- edit dns software configuration file

[inside dnsmasq.conf]

interface=lo
bind-interfaces
listen-address 127.0.0.1
cache-size=1000
resolv-file=/etc/resolv.dnsmasq
no-poll

%vim /etc/resolv.dnsmasq

[inside resolv.dnsmasq]

nameserver 192.168.30.1

%vim /etc/resolv.conf

[inside resolv.conf]

#comment out both existing nameservers
#comment out both existing nameservers

nameserver 127.0.0.1

% systemctl status dnsmasq
- checks for dnsmasq service activity

% systemctl enable dnsmasq
- enables automatic operation

% systemctl start dnsmasq
- starts operation

% nslookup google.com

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- Resource Record Types(Windows Server DNS Tool)

- forward & reverse lookup zones

- fwd: corp.515support.com
**dc10.corp.515support.com -> 10.1.16.1
- above is the lookup of the "A record" associated with the host name "dc10.corp.515support.com"

- rev: 1.16.1.10.in-addr-arpa -> dc10.corp.515.support.com
- above is the lookup of the hostname associated with the pointer record "1.16.1.10.in-addr.arpa"
**ipv6 appends "ip6.arpa" to the reverse ipv6 quartets for reverse lookup

- stored information

- start of authority(SOA)
- identifies the primary authoritative name server that maintains complete resource records for the zone

- name server(NS)
- identifies authoritative DNS name servers for the zone
- secondary name servers are for load balancing

- A records: ipv4
**host records
- ipv4 association with hostname

- AAAA records: ipv6
**host records
- ipv6 association with hostname

- pointer records(PTR)
- information stored in a reverse lookup zone

- mail exchange(MX)
- email server

- service record(SRV)
- enables network clients to locate domain controllers and related services(kerberos, ldap, etc)

- cname(canonical name) records
- aliases for existing records

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packetFiltering(macbook firewall)

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%/etc/pf.conf
- location of default pf configuration file

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%sudo nano /etc/pf.anchors/block_mDNS

{inside the file}
#Block mDNS (Bonjour) multicast traffic
block drop out quick proto udp from any to { 224.0.0.251, ff02::fb } port 5353
- creates file containing deny rule for multicast egress traffic from port 5353

%sudo nano /etc/pf.conf

{inside the file}
anchor "block_mDNS"
load anchor ""block_mDNS" from "/etc/pf.anchors/block_mDNS"
- "loads rule block_mDNS" into the config file

%sudo pfctl -f /etc/pf.conf
%sudo pfctl -e
- loading the new configuration and enabling packet filtering

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%sudo tcpdump -i en0 udp port 5353
- tcp traffic to/from socket en0:5353

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Windows Terminal

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>hostname
- displays current system name

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>help copy
- displays reference for command “copy

>dir
- displays list of directories in current file path

>fsutil fsinfo drives
- lists available drives on the current system “”c:\ e:\ etc”

>w:
- moves to specified drive letter

>vol w:
- displays basic information about drive “w”

>cd/mkdir/rmdir
- same as zshell/bash

>C:\Users\professor
- file path beginning with the C: drive letter
**file paths to other drives, devices, or hosted drives require their respective file paths
H:\cloud\yaMean

>diskpart
- DON'T USE THIS

>format k:
- formats drive letter “k:”
- DON'T USE THIS - it erases then partitions the drive letter

++

>copy /v P3K-447-report.log e:
- prompts user and copies file “P3K-447-report.log” to file letter e:

>copy P3K-447-report.txt duplicate.txt
- copies document “P3K-447-report.txt” to “duplicate.txt”

>xcopy /s Documents m:\backups
- copies directory “Documents” and its contents to file path “m:\backups”

>robocopy
- copies files with extensive features
**more robust copy

++

>winver
- displays version of windows

>gpupdate
- updates group policy

>gpupdate /target:{computer|user} /force
- forces computer or user group policy update

>gpresult
- displays verification of group policy setting for the current user or computer

>gpresult /r
- compiles all information about current gp settings on the machine

>gpresult /user sgc/professor /v

>net view \\sgc-ad
- displays available shares on server “sgc-ad”

>net view /workgroup:<workgroupname>
- displays available shares in the specified workgroup

>net use w: \\sgc-ad\Classified
- maps the specified network share “Classified” to the drive letter “w”

>net user <username>
- displays users on a specified device

>net user <username> * /domain
- displays users in the specified domain

++

>pathping -n 9.9.9.9
- combines traceroute and metrics about the path
- [-n] halts DNS resolution

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>w32tm
- command associated with configuring the NTP

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Install Locations

- android sdk: /Users/dunnkymani/Library/Android/sdk
- android projects: /Users/dunnkymani/AndroidStudioProjects

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Windows Terminal

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windows key + s
- open command prompt

ipconfig /release
- clears every interface on the computer of its address assignment
**ipconfig /release “Wi-fi”
- specifies the interface named “Wi-fi”

ipconfig /displaydns | clip
- copies output of dns cache
- pasteable to text editor software

getmac /v
- all computers mac addresses

powercfg /energy
- power report in reference to computer components
**copy output file location and enter into commands prompt

powercfg /batteryreport
- power report in reference to battery usage

assoc
- displays file type -> software associations

assoc .mp4=Vlc.vlc
- changes file type -> software association

chkdsk /f
- “check disk” for errors

sfc /scannow
- system scan

DISM /Online /Cleanup-Image /CheckHealth
- system scan and health check

DISM /Online /Cleanup-Image /ScanHealth
- longer scan

DISM /Online /Cleanup-Image /RestoreHealth
- system restoration

tasklist
- displays running tasks on computer

tasklist | findstr script
- displays running task with “script” in the name

taskkill /f /pid
- /f -> force
- pid selects the specific task by process identity

netsh wlan show wlan report
- generates wireless lan report
- copy file path to another line to open the report

netsh interface show interface
- displays interfaces

netsh interface ip show address | findstr “IP Address”
- dislplays IP addresses

netsh interface ip show dnsservers
- displays dns servers

netsh advfirewall set allprofiles state off
- disables windows firewall
- …on to turn it back on

netstat -af
- shows all connections and listening ports
- displays FQDNs instead of just IPs

netstat -o
- shows all active TCP connections as well as the PID

tasklist /FI “PID eq 1234”
- collectively the statement generates the software holding the connection
- lists running processes
- applies a filter to limit the output
- filter condition enclosed in the quotes

netstat -e -t 5
- [-e]displays ethernet statistics
- [-t 5] repeats the display every 5 seconds

netstat -s
- show detailed stats per protocol(TCP, UDP, ICMP, etc)

netstat -aon
- show all active TCP/UDP connections and their PIDs

route print
- displays routing table for the computer

route add 192.168.40.0 mask 255.255.255.0 10.7.1.44
- adds static route to computer table

route delete 192.168.40.0
- removes entry from the table

shutdown /r /fw /f /t 0
- [/r] restarts the computer after shutdown
- [/fw] enters firmware(UEFI/BIOS) settings automatically
- [/f] force-closes all applications without saving
- [/t 0] sets timer to 0 seconds

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Disk Utility(ZShell)

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%diskutil list
- lists connected volumes

%diskutil info -all
- verbose information

%diskutil eject “/Volumes/VolumeName”
- ejects and powers down the device “VolumeName”

%diskutil unmount force “/Volumes/VolumeName”
- forcefully unmounts the device

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Android Debug Bridge(android shell)

++

- install android debug bridge

- mash build number to activate developer mode

- developer options

- enable usb debugging

- ADB commands

%adb devices
- lists android devices

%adb -s emulator-5554 shell
**specific device string must be appended when there's multiple
- opens device shell(phone cli)

%adb shell getprop
- returns all properties of the phone is unspecified

++

Troubleshooting the script

%adb shell

%touch temp.txt
- creates a document

%echo "hello" > temp.txt
- saves string to temp.txt document

%cat temp.txt
- displays contents of target file(hello)

%adb push pc_phone.txt /sdcard
- file transfer from pc to phone's sd card

%adb pull /sdcard/pc_phone.txt pc/
- file transfer from phone to pc directory

++

%adb shell input tap x y 538 1099
- simulates tape on the screen

- settings

- developer options

- pointer location
**observe the coordinates for the "taps"

%adb shell input text "hello"
- simulates input of hello on the keyboard

%adb shell input keyevent 4
- simulates phone button depress

++

%adb shell list packages
- lists apps installed

%adb shell list packages I grep bk
- lists apps containing "bk"

++

%adb shell

%run-as bk.yt.example
- basically to the directory the app "bk.yt.example" stores information
**pwd to check

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++
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Sudo Code & Scripting

++

start loop

%adb -s 6fb1d29d shell input swipe 45.0 935.0 685.0 935.0
#swipe left to right

%adb -s 6fb1d29d shell input tap 59.0 116.0
#tap 3 lines top left

%adb -s 6fb1d29d shell input tap 281.0 116.0
#taps sign in/sign up

%adb -s 6fb1d29d shell input text

%adb -s 6fb1d29d shell input tap 388.0 669.0
#taps away error message

repeat

**commands are represented with their preceeding delimiter(config#; %)
**scripts uses, but with different context

++

NetworkChuck Bash Scripting

which $SHELL
- displays the current working shell

echo "Hi Mom"
- displays Hi Mom

nano himom.sh
- creates an executable text file, "himom"

[[himom.sh]]

#!/bin/bash
- declares the shell in use for the script

echo "Hi Mom"

++

[[inside bestdayever.sh]]

#!/bin/bash

echo "Good Morning Chuck!!"

sleep 1
- pauses actions for 1 second

echo “You’re looking good today Chuck!!”

sleep 1

echo “You have the best beard I’ve ever seen Chuck!!”

++

[[bestdayever.sh]]: using variables

#!/bin/bash

name “Patricia”
- declares a variable “name”, as “Patricia”

echo “Good Morning $name!!”
- $name: utilizes the variable
sleep 1
- pauses actions for 1 second

echo “You’re looking good today $name!!”

sleep 1

echo “You have the best beard I’ve ever seen $name!!”

++

[[bestdayever.sh]]: read(accepting input)

#!/bin/bash

echo “What is your name”

read name
- accepts input

echo “Good Morning $name!!”

sleep 1

echo “You’re looking good today $name!!”

sleep 1

echo “You have the best beard I’ve ever seen $name!!”

++

[[bestdayever.sh]]: read(using positional arguments[parameters])

#!/bin/bash

name = $1

echo “Good Morning $name!!”

sleep 1

echo “You’re looking good today $name!!”

sleep 1

echo “You have the best beard I’ve ever seen $name!!”

%./bestdayever.sh Ben
- executes script, using “Ben” as the 1st positional parameter

[[bestdayever.sh]]: read(using positional arguments[parameters])

#!/bin/bash

name = $1

compliment = $2

echo “Good Morning $name!!”

sleep 1

echo “You’re looking good today $name!!”

sleep 1

echo “You have the best $compliment I’ve ever seen $name!!”

%./bestdayever.sh Elitha Eyes
- executes script, using “Elitha” as 1st, and “Eyes” as the second

++

[[bestdayever.sh]]: read(using positional arguments[parameters])

#!/bin/bash

name = $1
compliment = $2

user=$(whoami)
- declares “user” as the output of a command
date=$(date)
whereami=$(pwd)

echo “Good Morning $name!!”

sleep 1

echo “You’re looking good today $name!!”

sleep 1

echo “You have the best $compliment I’ve ever seen $name!!”

sleep 2

echo “You are currently logged in as $user and you are in the directory $whereami. Today is $date”
- displays message including 3 variables generated from commands %whoami, %date, and %pwd

%./bestdayever.sh Elitha Eyes

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xxd(utility)

%echo “In hexidecimal” | xxd -p
- prints a string of hexidecimal digits representing the phrase ‘In hexidecimal’

%echo “0123456789abcdef” | xxd -r -p
- reverses and decodes hexidecimal string into plain text

++

%cat data.txt
- displays plain text information contained inside file “data”
**assume the information is a hexdump that needs decoding

%xxd -r data.txt data
- reverses the hexdump into machine language
- removes the current file type and extention .txt

- file signature(magic numbers/bytes) are inserted in the beginning of a file
**00000000: 1f8b 0808

- search list of file signatures for the matching bytes
**1f 8b -> GZIP compressed file -> .gz

%mv data data.gz
- change the file name to a .gz(file extension of the matching hexdump "head")

%gzip -d data.gz
- uncompresses the file "data"
**be sure to use the corresponding technology for each file signature/type

%cat data
- displays decoded information

%xxd data
- displays a hexdump of the file contents inside "data"

- repeat the cycle from finding the file signature until hexdump is fully decoded

%xxd data | head
- hexdump limiting the output to the first few lines using pipe and the phrase "head"

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File

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file ./Example.txt
- displays file type of "Example.txt"

++
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Link(sysmink)

++

% ln -sv ./test1/test1.txt
- creates shortcut in current directory leading to file "test1.txt"

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++

Remove(rm)

++

%rm -Riv Downloads/docs/blcPacket
- removes directory "blcPacket"
- removes its contents(-R)
- prompts user before every removal(-i)
- informs user of actions(-v)

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Copy(cp)

++

%cp -Rv /path/to/source_directory /path/to/destination_directory
- copies "source_directory" to "destination_directory"
- copies contents as well(-R)
- lists the actions taken(-v)

++
++

SCP(sftp via ssh)

++

%scp -P 22 ./local.txt root@172.16.0.2:/root
**scp [localPath] [user@]hostIP:[remotePath]
- transfers local file named "local.txt" to "remotePath" on remote system "user" at address "hostIP"
- [-P] specifies port #

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Netcat

++

! target device

% nc -l 1337
- netcat listening on port 1337

! sender

% nc 192.168.1.132 1337

% Hello?
- distant end displays "Hello?"

! port scans address for open port, every 1 second within 1-1000 range whilst disregarding I/0 latency(-z)

% nc -v -w 1 10.73.31.9 -z 1-1000

++

! target device

% nc -l -p 1234 > winServ.iso
- listening (-l)
- port 1234 (-p)
- downloads throughput to file "winServ.iso"
**file names don't have to be consistent amongst terminals

! sender

% dd if=winServ.iso bs=1M status=progress | ncat 172.16.0.2 1234 -q 1 < winServ.iso
- outputs real-time transfer metrics for the ncat command
- quits 1s after file transfer(-q)
- uploads file "winServ.iso"

++
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GPG

++

%gpg --list-keys
- lists all public and private keys saved locally

%gpg --full-generate-key
- creates asymmetric key pair with editable properties such as encryption standard, expiration, credential associations

%gpg --edit-key me@example.com
- edits key pair associated with email: me@example.com
**properties located in gpg manual page

%gpg --encrypt data.txt
- generates encyption prompt for “data.txt” using asymmetric(recipient) key pair of choice

%gpg --encrypt --recipient 123456789ABCDEFGHIJKL data.txt
- encrypts “data.txt” using public key from pair associated with uid “123456789ABCDEFGHIJKL”

%gpg --decrypt data.txt.gpg
- generates decryption prompt regarding “data.txt.gpg” for the private key passphrase of choice
- displays data on screen

%gpg -- decrypt data.txt.gpg > decryptedData.txt
- decrypts “data.txt.gpg” and generates file containing unencrypted file contents by the name of “decryptedData.txt”

%gpg -- export 123456789ABCDEFGHIJKL > public-key.gpg
- outputs uid “123456789ABCDEFGHIJKL” public key to file “public-key.gpg”

%gpg -- export --armor 123456789ABCDEFGHIJKL > public-key.gpg
- outputs uid “123456789ABCDEFGHIJKL” public key in human readable format to file “public-key.gpg”
**%mv public-key.gpg public-key.asc
- changes format of public key file to universal text file with import capabilities(ascii)

%gpg -- import public-key.asc
- imports public key contained within file “public-key.asc”
- simply copy public key from the source(website, email, etc) and import to gpg

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Active Directory Hierarchy

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Typical uses & notes

- GPOs are linked to domains, OUs, or sites — OUs are commonly used to target GPOs to sets of users or computers
- Delegation: Use OUs to delegate administrative permissions without giving domain-level rights
- Trusts: Forests may contain multiple trees which implicitly trust each other inside the forest; cross-forest trusts can be created between different forests
- Global Catalog: Stores a partial replica of all objects for the forest to facilitate searches across domains

++

Namespace / DN examples

- DNS domain name: corp.contoso.com
- Lightweight Directory Access Protocol(LDAP) Distinguished Name (DN): CN=Alice,OU=Users,DC=corp,DC=contoso,DC=com
- UserPrincipalName (UPN): alice@corp.contoso.com (often used to sign in)

++

Key ideas (short)

- Forest: Top-level security boundary. Contains one or more trees. Example: contoso.com
- Tree: A contiguous namespace collection of domains that share a hierarchical DNS name (e.g., contoso.com and eng.contoso.com). Trees inside a forest share the Global Catalog and schema
- Domain: Administrative and replication boundary. Has a DNS name (e.g., corp.contoso.com)
- Organizational Unit (OU): Container used to group objects for delegation, GPO application, and organization (e.g., OU=Users)
**generic containers(NOT TO BE CONFUSED WITH OUs): store system objects/parent objects of the created containers
- reject GPO application and other management differences from OUs
- Objects: Users, Computers, Groups, Contacts, etc. Each object has attributes and a unique Distinguished Name (DN)

++

```
flowchart TD
    subgraph Forest[Forest: contoso.com]
        direction TB
        subgraph TreeA[Tree: contoso.com]
            direction TB
            subgraph Domain1[Domain: corp.contoso.com]
                direction LR
                OU1[OU: Users]
                OU2[OU: Servers]
                OU3[OU: SecurityGroups]
                OU1 --> U1[User: Alice]
                OU1 --> U2[User: Bob]
                OU2 --> C1[Computer: SRV01]
                OU3 --> G1[Group: Finance-Admins]
            end
            subgraph Domain2[Domain: eng.contoso.com]
                direction LR
                OU4[OU: Dev]
                OU5[OU: Test]
                OU4 --> U3[User: Carol]
                OU5 --> C2[Computer: TEST01]
            end
        end

        subgraph TreeB[Tree: partner.contoso.net]
            direction TB
            subgraph Domain3[Domain: partner.contoso.net]
                OU6[OU: ExternalUsers]
                OU6 --> U4[User: vendor1]
            end
        end
    end
```

++
++

Protocols, Properties, and Parameters = Tactics, Techniques, and Procedures
- an inference developed from my own experience on the hierarchy of command line technologies, purposes, and concepts

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Virtual Tunnel Interface(VTI)

++

```
(config)# interface Tunnel1
(config-if)# ip address 10.0.0.1 255.255.255.252
(config-if)# tunnel source GigabitEthernet1
(config-if)# tunnel mode ipsec ipv4
(config-if)# tunnel destination 203.0.113.2
(config-if)# tunnel protection ipsec profile VTI-IPSEC-PROFILE
```

++
++

IKEv2 Point-to-Point VPN

++

```
HQ-ROUTER#int g0/0
    #ip address 192.0.2.2 255.255.255.252
```

```
#crypto ikev2 keyring Keys1
#peer Branch-Rtr
#address 209.165.201.2
#pre-shared-key local HQ-Keystring
#pre-shared-key remote Branch-Keystring

#crypto ikev2 profile default

#match identity remote fqdn Branch-Rtr.abc.public
#identity local fqdn HQ-Rtr.abc.public
#authentication local pre-share
#authentication remote pre-share
#keyring local Keys1


BRANCH-ROUTER#int g0/0
#ip address 209.165.201.2 255.255.255.252

#crypto ikev2 keyring KR1
#peer HQ-Rtr
#address 192.0.2.2
#pre-shared-key local Branch-Keystring
#pre-shared-key remote HQ-KeyString

#crypto ikev2 profile default

#match identity remote fqdn HQ-Rtr.abc.public
#identity local fqdn Branch-Rtr.abc.public
#authentication local pre-share
#authentication remote pre-share
#keyring local KR1
```

++
++

Dot1x Access Control

++

LOCAL MAB ONLY

```
#aaa new-model

#aaa authentication dot1x default local

#interface g1/0/1
#switchport mode access
#authentication port-control auto
#mab

#username <MAC> privilege 0 secret <MAC>
```

++

Dot1x + MAB

```
#radius server exampleRadius
#address ipv4 1.1.1.1 auth-port 1812 acct-port 1813
#key localMAB

#ip radius source-interface vlan1000

#aaa group server radius exampleGroup
#server name exampleRadius
#ip radius source-interface vlan1000

#dot1x system-auth-control

#aaa new-model
#aaa authentication dot1x default group radius local
#aaa authorization network default group radius local

#int g1/0/1
#switchport mode access
#authentication host-mode multi-domain
#authentication closed
#authentication port-control auto
#mab
#dot1x pae authenticator

#username <MAC> privilege 0 secret <MAC>

#show authentication sessions interface
#show radius statistics
```

++
++

Flexible Netflow

++

```
#flow exporter EXPORTER1
    ! NetFlow transmitter config
#destination 10.1.1.100
#source loopback 0
#transport udp 2055

#flow record LANFLOW1
    ! Determines what NetFlow captures(match) and displays(collect)
#match ipv4 tos
#match ipv4 protocol
#match ipv4 source address
#match ipv4 destination address
#match transport source-port
#match transport destination-port
#match interface input
#collect routing next-hop address ipv4
#collect interface input
#collect routing next-hop address ipv4
#collect interface output
#collect counter bytes
#collect counter packets
#collect timestamp sys-uptime first
#collect timestamp sys-uptime last

#flow monitor MONITOR1
    ! Comprehensive NetFlow construct
#description Primary NetFlow Cache
#exporter EXPORTER1
#record LANFLOW1

#int f0/0
#ip flow monitor MONITOR1 input

#int f0/1
#ip flow monitor MONITOR1 input

#show flow monitor
#show flow monitor MONITOR1 cache
```

++

Basic Netflow

++

```
#interface f0/0

#ip flow ingress
    - enables WAN -> LAN netflow monitoring
```

#ip flow egress
- enables LAN -> WAN netflow monitoring

#ip flow-export version 9
- changes default netflow version 1 to version 9

#ip flow-export destination 1.2.3.4 9999
- exports netflow messages to udp socket 1.2.3.4:9999

#ip flow-export destination 4.3.2.1 8888 sctp
- exports netflow messages to TCP socket 4.3.2.1:8888
- uses Stream Control Transmission Protocol

#show ip flow interface
#show ip flow-export
#show ip cache flow

++
++

Logging Messages(syslog, debugging for Cisco)

- displays current logging settings
#show logging

- enables console logging
#logging console

- enables console logging for message entry and higher
#logging console <message level>

- sends syslog messages to VTY connected terminals
#logging monitor
**disabled by default

- enables syslog message display when connected via VTY lines
#terminal monitor
**disabled by default

- sends syslog messages to the buffer
#logging buffer

- sends syslog messages to the buffer, determines buffer size, and message level
#logging buffer 8192 6

- sends syslog messages to an external server
#logging 192.168.1.1
#logging trap <level>
**determines syslog level for messages sent to external server

- ensures messages don't interrupt user input
#logging synchronous

- displays current debug messages enabled
#show debug

- disable logging messages
#no logging console
**enabled by default(disable when not connected via console line)
#no logging monitor

- disable debug conditions
#undebug < debug condition>
#undebug all
**disables all messages

++
++

SNMP

++

R1#snmp-server community Public ro
- enables a snmp community "Public" that is read-only
R1#snmp-server host 10.10.10.250 Public
- configures router to send SNMP messages to host "10.10.10.250"
- authenticating with the community string "Public"

++
++

NTP

++

- authentication
**authentication parameters/properties must be identical across systems participating the in the NTP relationship

R1#ntp authenticate
- enables authentication to ensure device syncs to intended NTP server only

R1#ntp authentication-key 1 md5 jeremysitlab
- creates authentication hash "1" protected by password "jeremysitlab"

R1#ntp trusted-key 1
- lists key "1" as trusted and available for use

R2#ntp authenticate
R2#ntp authentication-key 1 md5 jeremysitlab
R2#ntp trusted-key 1
R2#ntp server 10.0.12.1 key 1
R2#ntp peer 10.0.23.2 key 1
- adds authentication to the peer relationship with R3

R3#ntp authenticate
R3#ntp authentication-key 1 md5 jeremysitlab
R3#ntp trusted-key 1
R3#ntp server 10.0.12.1 key 1
R3#ntp peer 10.0.23.1 key 1

++

R1#ntp master
- sets local loopback address as the ntp server(top level server)
- default stratum is 8
- operates without a primary or secondary server to sync to

R2#ntp server 10.0.12.1
R2#ntp peer 10.0.23.2

R3#ntp server 10.0.12.1
R3#ntp peer 10.0.23.1

++

R1# int lo0
R1#ip address 10.1.1.1 255.255.255.255
R1#exit
R1#ntp source loopback0
- ntp messages will originate from the loopback address

R2#int lo0
R2#ip address 10.2.2.2 255.255.255.255
R2#ntp server 10.1.1.1
R2#show ntp associations

R3#ntp server 10.1.1.1
R3#ntp server 10.2.2.2

++

R1#ntp server 216.239.35.0 prefer
R1#ntp server 216.239.35.4
R1#ntp server 216.239.35.8
R1#ntp server 216.239.35.12
- configures four ntp servers on router 1, setting .0 as the preferred server

R1#show ntp associations
- *: currently synced ntp
- +: ntp candidate
- ~: configured

R1#show ntp status
R1#show clock detail
- NTP only uses UTC
- time zone must be configured
R1#show calendar
- NTP does not update your calendar
- calendar must be configured to sync with the ntp server

R1#clock timezone JST 9

R1#ntp update-calendar

++

- NTP provides a solution for the lack of scalability in terms of network synchroniztion

- NTP is a client/server architecture in which the participating devices can be both(with an additional symmetric active mode)

- NTP operates on UDP 123

- reference clocks
- atomic, GPS clocks and the like -> very accurate
- considered stratum 0

- primary servers
- servers with direct connection to the reference
- considered stratum 1

- secondary servers
- receiving timing from other NTP servers
- considered stratum 2 and above

- NTP peering(symmetric active mode)
- devices at the same stratum level syncing with one another for accuracy and redundancy

++

#show clock
- displays software time/date
- default time zone is UTC(coordinated universal time)

#show clock detail
- displays software time/date more detailed
- * -> means time is not authoritative

- NTP alleviates hardware calendar drift, and provides synchronization for accurate logs used for troubleshooting

#show logging

#clock set 14:30:00 27 DEC 2020
- manual clock configuration protocol for the SOFTWARE CLOCK

#calendar set 14:35:00 27 DEC 2020
- manual calendar configuration for the HARDWARE CLOCK

#clock update-calendar
- syncs the calendar to the clock

#clock read-calendar
- syncs the clock to the calendar

#clock timezone JST 9
- "JST" is a word in this instance(not a property), the engineer uses to represent Japan Standard Time
- "9" sets the clock 9 hours ahead of UTC

#clock summer-time EDT recurring 2 Sunday March 02:00 1 Sunday November 02:00
- manually configures daylight savings start and stop

++
++

GK

- broadcast = NetID w/ host bits inverted
ex. 127.0.0.0/8 -> 01111111.0.0.0 -> 01111111.11111111.11111111.11111111 -> 127.255.255.255

- ip range 127.0.0.0/8: loopback addresses
- aka localhost
- aka DNS: "hostname"

- set aside ranges
- 0.0.0.0/8
- 255.255.255.255
- 64.0.0.0/10, 192.0.0.0.24, 192.88.99.0/24, 198.18.0.0/15
- 0.2.0.0/24, 198.51.100.0/24, 203.0.113.0/24

++
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Powershell Network Cmdlts

powershell>Get-NetAdapter
powershell>Get-NetIPAddress
- queries the existing configuration

powershell>New-NetIPAddress
- new configuration

powershell>Set-NetIPAddress
- modifies existing configuration

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>netsh interface ip set address "Ethernet" dhcp
- sets interface "Ethernet" to accept ip address using dhcp

>netsh interface ip set address "Ethernet" static 198.51.100.17 255.255.255.240
- sets interface "Ethernet" to static address

>netsh interface ip show config
- displays the current ip configuration

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QoS Config

- class map: identify traffic types

#class-map HTTPS_MAP
#match protocol https
#exit

#class-map HTTP_MAP
#match protocol http
#exit

#class-map ICMP_MAP
#match protocol ICMP
#exit

- policy map: actions to be taken

#policy-map g0/0/0_OUT

#class HTTPS_MAP
#set ip dscp af31
#priority percent 10
#exit

#class HTTP_MAP
#set ip dscp af32
#bandwidth percent 10
#exit

#class ICMP_MAP
#set ip dscp cs2
#bandwidth percent 5
#exit
#exit

- service policy: applies QoS setting to particular interface

#int g0/0/0
#service-policy output g0/0/0_OUT

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Quality of Service(QoS)

- applies various treatments to different traffic

ex. bandwidth: 20% voice•30% specific•50% other

- delay
  - one-way: time required from source -> destination
  - two-way: time required from source -> destination + destination -> source

- jitter: variation in one-way delay between packets sent by the same application
  - jitter buffer provides fixed delay for packets, such as audio

- loss: purposefully discarding packets in the event queues get full

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- Audio Call Standards

  - OWD: x =< 150 ms

  - Jitter: x =< 30ms

  - Loss x =< 1%

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- Traffic Queuing

  - queued messages are forwarded in first-in-first-out(FIFO) method traditionally
  - tail dropped: new packets are discarded onced the queue is full
    - TCP Global Sync: perpetual throttling and amplification cycle caused by excessive tail drops
    ex. network congestion -> tail drops -> Global TCP Sync -> window decrease ->
    >network underutilization -> window increase -> network congestion…

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- Solution: Tail Dropping

  - random early detection(RED): purposely drops packs when traffic reaches a user defined threshold
    - regulates the TCP window

  - weighted random early detection: similiar to RED w/ granular packet controls defined by traffic classifications

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- VoIP: voice over IP

- PSTN: Public switched telephone network
  - aka: plain old telephone service(POTS)

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- IP Phones

  - 3-port switch(traffic -> phone -> switch)
  - uplink: host -> external network
  - downlink: external network -> host

  - Data & Voice VLANs
    **links carrying both are NOT trunks

    #int g0/0
    #switchport mode access
    #switchport access vlan10
      - this is the data vlan
    #switchport voice 11
      - enabling voice vlan on the link[the link is still an access port

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- PoE: Power Over Ethernet

  - power sourcing equipment(PSE - provider)
  - powered devices(PD - recipient)
    **AC -> PSE -> DC -> PD

  #power inline police(#power inline police action err-disable)
    - disables port; sends syslog message during PD overconsumption

  #power inline police action log
    - restarts interface and sends suslog in the event of over consumption
    - PD loses power temporarily

  #show power inline police g0/0
    - displays available, used, remaining power, and interface power settings

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- Classification/Marking

  - traffic is organized into classes

- methods
    - ACL: similiar parameters as NAT usage
    - Network Based Application Recognition(NBAR): deep packet inspection
    - Priority Code Point(PCP)/Class of Service(CoS)
        - ONLY OPERABLE over trunk or access + voice links
        - Layer 2 Classification(3 bits)
            - 0: best effort or "best effort…of whats left"
            - 1: background
            - 2: excellent effort
            - 3: critical applications -> IP phones call signaling(PCP3)
            - 4: video
            - 5: voice data -> IP Phone(PCP5)
            - 6: Internetwork Control
            - 7: Network Control
    - Differentiated Services Code Point(DSCP)/Type of Service(ToS)
        - developed as a 3 bit extention to the IPP standard, a layer 3 equivalent to PCP
        - Layer 3 Classification(6 bits)
            - 0: Default Forwarding(DF) -> best effort
            - 46: Expedited Effort(EF) -> low loss/latency/jitter traffic(voice)
            - Assured Forwarding(AF) -> 12 values; 4 classes; 3 drop precedences
                - syntax: AFXY -> X = class Y = drop precedence
                - Range: AF13 - AF41 -> [4] highest class [1] least likely to be dropped
                    ex. least to priority: AF13 -> AF12 -> AF11 -> AF21 -> AF31 -> AF41
                - Binary: XY = _ _ _ _ ll _ _ _ ll 0
                    ex. AF11 = 001010 -> 0 0 1 ll 0 1 ll 0 -> 1 ll 1 -> AF11
                    **3 digits class - 2 digits drop precedence - last digit is always 0
            - Class Selector(CS) -> set of 8 standard values; backwards compatible with IPP
                - syntax: CS# -> # = 0-7
                - Range: CS0 - CS7
                - Binary: # = _ _ _ _ ll 0 0 0
                    ex. CS3 = 011000 -> 0 1 1 ll 0 0 0 -> 0 1 1 -> CS3
                    **3 digits CS - last 3 digits are always 0
            - **DSCP representation(DSCP10, DSCP24, etc) are simply the decimal equivalent of the entire binary number
                ex. AF11 -> 001010 -> DSCP10
                ex. CS3 -> 011000 -> DSCP24
        - DSCP Recommendations
            - Voice Traffic: EF
            - Interactive Video: AF4x
            - Streaming Video: AF3x
            - High Priority Data: AF2x
            - Best Effort: DF

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- Trust Boundaries
    - point at which devices trust/untrust traffic QoS markings
        - trusted: forwarded without changes
        - untrusted: markings changed according to policy

    EF -> DF ->
    Cos0 CoS0

    PC1 -> PH1 -> *SW1* -> R1 -> R2 -> Internet
    PC2 -> PH2 -> *SW2*

    Trust Boundary: traffic prior to the switches will be changed

    EF -> EF ->
    Cos0 CoS0

    PC1 -> *PH1* -> SW1 -> R1 -> R2 -> Internet
    PC2 -> *PH2* -> SW2

    EF
    CoS0 -> DF

    Trust Boundary: better placements of the TB to ensure voice traffic retains DSCP classificartion

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- Queuing/Congestion
    - RED and WRED drop packets early to avoid tail drop
        - faster rate traffic is queued, and dropped once the queue is full
    - devices match traffic based on various factors(DSCP header) then queue
    - interfaces are limited to one frame at a time
        - scheduler determines next frame
        - traffic is scheduled based on priority

    ingress -> routing,etc -> classification -> queuing -> scheduling -> transmission

    - Schduler Methods
        - weighted round robin
        - Class Based Weighted Fair Queuing(CBWFQ)
            - functions similiarly to weighted round robin with added bandwidth reservation
            - not ideal for voice data as it adds delay and jitter

            congestion -> routing -> 10% classA -> queueA -> scheduler -> transmission
                                     15% classB -> queueB ->
                                     15% classX -> queueX ->
                                     60% classY -> queueY ->

            - schduler: YXBA…

- Low Latency Queue(LLQ)

    - designates one or more queues as strict priority queues
    - scheduler always takes next packet from priority queue until its empty

    congestion -> routing -> 10% classA -> queueA -> scheduler -> transmission
    15% classB -> strictPriorityQueue ->
    15% classX -> queueX ->
    60% classY -> queueY ->

        - scheduler: BYXA…
        - reduces delay and jitter of voice/video
        - potentially starves other queues

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- Shaping & Policing

    - shaping: buffers traffic in the queue if configured rate is exceeded

    - policing: drops traffic in the queue if configured rate is exceeded
        - offers marking instead of dropping
        - burst traffic: allows overages for a short period of time
            - traffic is sent in bursts instead of a constant stream

    SW1 -> CusR -> IspR ->Internet

        - customer pays for a 300mpbs package
        - customer router [shapes] oubound traffic to 300mbps
        - ISP router [polices] traffic to 300mbps

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Zsh(Apple Commands)

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- % mv -f test.txt test1.txt
    - renames and overwrites file “test.txt” to “test1.txt”

- % mv Documents/test.txt Documents/directory/
    - moves file “test.txt” in directory “Documents” to directory “directory” inside of “Documents

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- % open -R Documents
    - reveals directory “Documents” in finder

- % open -a TextEdit test.txt
    - opens documents “Test.txt” with application TextEdit

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- % find -d Documents
    - lists directory “Documents” contents recursively from deepest to highest

- % find -s Documents
    - lists directory “Documents” contents recursively in alphabetical order

- % find ~ -iname “*abc*”
    - searches home directory for filename containing abc

- % find . -size 33c -exec ls -al {} +
    - searches current directory and returns any file 33 bytes in size
    - [-exec] built-in direction telling find to execute list command afterwards (proper syntax != pipe)
    - {} place-holder for arguments to be passed to list command
    - [+] tells find to run list command once all files are found

- % find / -user bandit7 -group bandit6 -size 33c 2>/dev/null
    - searches root and returns file
    - [-user] owned by user “bandit7”
    - [-group] owned by group bandit6
    - [2>/dev/null] suppresses “permission denied” returns

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- % nmap -sP x.x.x.x/x
    - ping network to reveal hosts

- % nmap -sT -p 80,443 x.x.x.x/x
    - full open scan for network devices with open http, https ports

- % nmap -sS -p 80,443 x.x.x.x/x
    - stealthy scan for network devices with open http, https ports

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- % netstat -r
    - displays working routing table

- % netstat -I en0
    - displays network attributes of interface “en0”

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- % shutdown -r +10 “System will reboot”
    - initiates reboot and message display timer of 10m

- % shutdown -h +10
    - initiates halt and shutdown timer of 10m

- % shutdown -s +10
    - initiates a sleep timer of 10m

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- %top
    - displays detailed running processes
    **once in interactive mode, shift + ? for output manipulation help

- %top -s +10
    - displays running processes for 10 seconds

- %top -o command -s +10
    - displays running processes in alphabetical order for 10 seconds

- %top | grep Google\ Chrome
    - displays running processes containing the string “Google Chrome”

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- %ps -p 1234 -o pid,ppid,user,comm
    - displays PID PPID user and process path

- %ps -u dunnkymani | grep TextEdit
    - displays processes running under user “dunnkymani” including the string “TextEdit” in the name
    **prep TextEdit

- returns only PID

%kill 1234
- stops a process gracefully

%kill -9 1234
- force stops a process

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% ping -c 1000 127.0.0.1

% cntl + Z: sleeps running process

% jobs
- displays current running jobs

$ bg %ping\ -c\ 1000\ 127.0.0.1
- places job in background
**cntl + [c l z] will not work on background jobs

$ fg %ping\ -c\ 1000\ 127.0.0.1
- places job in foreground

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%ifconfig -l
- displays available network interfaces

%ifconfig en0 down
- shuts down interface “en0”

%ifconfig -L en0
- limits output to display network settings for interface “en0”

%ifconfig -X en
- displays network settings for interaces whose names include the regular expression “en”

% ifconfig en0 inet 192.0.2.10 netmask 255.255.255.0
- assigns 192.0.2.0/24 to interface “en0”

% ifconfig en0 inet 192.0.2.45/28 add
- adds 192.0.2.45/28 to addresses on interface “en0”

% ifconfig en0 inet 192.0.2.45 -alias
- removes 192.0.2.45 from interface “en0”

% ifconfig en0 inet6 2001:db8:bdbd::123 prefixlen 48 alias
- assigns IPv6 address w/ 48 bit subnet to interface “en0”

% ifconfig en0 inet6 2001:db8:bdbd::123/48 delete
- remove IPv6 address with 48 subnet from interface “en0”

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% aea encrypt -profile hkdf_sha256_aesctr_hmac__scrypt__none -i 1.txt -o one.aea -password-value -123
- generates encrypted “one.aea” file from “1.txt” protected by password “123”

% aea decrypt -i one.aea -o one.txt -password-value -123
- generates plain text file “one.txt” from encrypted file “one.aea” using password “123”

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% zip test.zip directory
- zips file named “directory” into a packge named test.zip

% zip -r test.zip directory
- zips file named “directory” into a package including its contents

% zip -er test.zip directory
- zips file named “directory” into a password protected package including its contents

% zip -erv test.zip directory
- zips file named “directory” into a password protected package, including its contents, whilst listing the actions

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% chmod +x private.txt
- adds execute permission for the user, group, and other

% chmod -x private.txt
- removes execute permission for the user, group, and other

% chmod u=w private.txt
- limits user to write permission
**% chmod u+r private.txt
- adds read permission explicitly for the user

% chmod 000 private.txt
- removes all permission for the user, group, and other

% chmod 777
- assigns permission to user, group, and other using rwx numerical associations

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% ls -l
- outputs detailed list of directory contents
**drwxrwxrwx = file permissions
- read, write, execute(open); “d” indicates the listing is a directory
- each “rwx” set refers to user, group, other respectively

- r = 4; w = 2; x = 1; 0 = no permission
ex. 777 = all permissions for user, group, other
ex. user: 4 2 1 = 7
group: 0 2 0 = 2
other: 4 2 0 = 6

% ls -ld
- prints file permissions for current working directory

% ls -ld ./example
- prints file permissions for directory “example” using relative path

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Terminal Help

** % bindkey
! displays shell bindings

** % bash
% help <command>
! switches to bash for cleaner help dialogue
! switch back to zsh afterwards

**tab completions

**WAY CLEANER HELP DIALOGUE: zip -h
- [-h] causes “zip” command options and syntax output

** -h OR --h OR --help
- simpler than manual pages

**tab completion: typing tab after any input lists command options
- ex. man + tab = man mandoc mandoc_soelim manpath

%[command]: issuing the command solely without following syntax will present available syntax

%man -k [search term]

- displays manual entries containing search term
**search term = an individual letter, returns a list of available commands

:/[search term]

- finds term entries within the manual

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1523 Settngs

- Time
- DD HH MM SS
**DD last two of Julian date
**HH MM SS(zulu[+5h])
- CLR & store synchronization

Modified Table of Organization and Equipment(MTOE)

Help Desk

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Ethernet Cables

- 568B: OW O GW B BW G BrW BR

- straight through(access -> host): 568B - 568B

- crossover(trunks): 568B -> 568B(inside out)

- rollover(cisco proprietary): 568b -> 568B(reversed)

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AAA - Authentication, Authorization, Accounting

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- verify the identity of a user, prevent unauthorized access, record user activity

- external servers: RADIUS & TACACS+
- RADIUS: remote authentication dial-in user service
- TACACS+: terminal access controller access-control system plus

- authentication methods: PAP & CHAP
- PAP: Password Authentication Protol - simple username and password authentication
- CHAP: Challenge Handshake Authentication Protocol - self explanatory, more secure authentication means

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- RADIUS

- Internet Engineering Task Force(IETF)-standard protocol

- authentication and authorization process combined

- encrypts passwords

- Authentication/Authorization: UDP port 1812

- Accounting: UDP port 1813

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- TACACS+

- Cisco-proprietary protocol

- separate authentication, authorization, accounting process

- encrypts the entire contents of packets

- AAA: TCP port 49

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#aaa new-model
- enables AAA services on the local device

**create local database user as a backup in the event you're locked out of your device

#aaa authentication ppp AAALab tacacs local
- configures AAA authentication on the router, active authentication method, and failover methods
**router uses PPP to authenticate username: "AAALAB" —> failover is the TACACS+ server —> failover is the local user database

#username AAALab password Cisco
- local user database entry credentials

P1R1#interface Serial0/1
#ppp authentication pap AAALab
#ppp pap sent-username P1R1 password cisco
- configures int S0/1 w/ PAP —> specifies the TACACS+ methods to use(AAALab)
- enables remote authentication to another device

P1R2#interface Serial0/0
#ppppp authentication pap
#ppp pap sent-username AAALab password cisco
- configures int S0/0 w/ PAP —> no TACACS+ methods
- enables remote authentication to another device

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Power Over Ethernet

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#show power inline
- displays available, used, remaining power, and port distribution

#power inline auto
- automatically detect and power inline devices

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DNS

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- dns is used to resolve human-readable names(google.com) to IP addresses

- machines such as PCs don't use names, they use addresses(ie. IPv4/IPv6)

- names are much easier for us to use and remember than IP addresses
  - what's the IP address of youtube.com

- when you type 'YouTube.com' into a web browser, your device will ask a DNS server for the IP address of youtube.com

- the DNS server(s) your device uses can be manually configured or learned via DHCP

  >ipconfig /all
    - displays NIC information including DNS address

  >ipconfig /all | findstr dns
    - similiar to cisco usage of pipe extensions
    - useful on any command
  >nslookup
    - DNS mode for lookup and DNS information

  >ipconfig /displaydns
    - DNS cache

  >ipconfig /flushdns
    - empties DNS cache

- DNS 'A' record: maps names to IPv4 addresses

- DNS 'AAAA' record: maps names to IPv6 addresses

- Port 53: DNS - UDP(messages less than 512 bytes); TCP(messages) greater than 512

\\..\Windows\System32\drivers\etc\hosts.txt
  - file containing mappings of IP addresses to host names

- routers can be configured as both DNS server(rare) and client

#ip dns server
  - configures router to act as the DNS server

#ip host R1 192.168.0.1
#ip host PC1 192.168.0.101
#ip host PC2 192.168.0.102
#ip host PC3 192.168.0.103
  - list of hostname/IP address mappings

#show hosts
  - displays router(servers) stored mappings

#ip name-server 8.8.8.8
  - DNS server current router will query when acting as the client

#ip domain lookup
  - enables current router to perform DNS queries
  **ip domain-lookup

#ip domain name
  - routers default domain name
  *ip domain-name

  - domain name is automatically appended to any hostname without a specified domain
  ex. ping PC1 -> ping pc1.<domain>.com

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FTP & TFTP

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- industry standard protocols used to transfer files over a network

- client - server topology

  - copies files to/from a server
  ex. upgrading network device OS

#show file systems
  - displays cisco storage mediums

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- TFTP: trivial file transfer protocol

  - file copy to/from the server

  - no authentication(user/PW)

  - no encryption

  - UDP port 69
    ex. Uploading to a server
    - client requests, server ACK messages
    - server resend ACK after a timer expires w/o data stream
    SIMPLY: i. connection(req & ack) ii. data transfer(data & ack) iii. connection termination(final data, final ack)

  #show version
    **displays running ios
  #show file system
    **displays on board storage
  #copy tftp: flash:
    **copy [source][destination
  ?192.168.1.1
    **server ip
  ?[file name on the server]
  ?[desired end state file name]
  #show flash
  #boot system [file path]
    **points to the file
  #write memory
    **saves file to nvram(non volatile)
  #reload
    **reboots
  #show version
  #delete [file path]
    **removes old ios
  #show flash

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- FTP: file transfer protocol

  - copies to/from server•navigate directories•add/remove directories•list files
    - tasks coincide with commands send to the server

- authentication required(user/PW)

- no encryption(including user/PW)

- TCP port 20 & 21
  - 21: control port(persistent connection•FTP commands•replies)
  - 20: data port(established and terminated as needed)
    - active mode: server initiates the TCP connection(client ACK)
    - passive mode: client initiates the TCP connection(server ACK)
      **TCP requests are typically dropped from a source to a destination protected by firewall
      **passive mode remedies this topology allowing the client to initiate the connection

#ip ftp username [username]
#ip ftp password [password]
#exit
#copy ftp: flash:
  **copy ftp://[ftp server ip]/[file]/[filename] flash:[filename]
  **copy ftp://[username]:[password]@[ftp server ip]/[file]/[]filename flash:[filename
    ^^bypasses default authorization configuration

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FRHP(First Hop Resolution Protocol)

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- 3 protocols

- primary objective is load balancing and network redundancy

- configures 2 or more routers in an active & standby topology using virtual gateway and MAC addresses

  - active/standby terms vary per FHRP protocol

- routers send multicast "hellos" to negotiate roles

- end devices send ARP requests for the vMAC assigned to the VIP

- active router responds with the vMAC OR the backup router sends gratuitous ARP responses to update MAC address tables

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- HSRP: hot standby router protocol

  - cisco proprietary

  - active/standby election

  - single active - per - subnet load balancing

  - v1/v2

    - v2 adds IPv6 support and increases the # of groups that can be configured

    - v1 & 2 are not compatible; configuration has to be consistent throughout the topology

  - v1Multicast: 224.0.0.2

  - v2Multicast: 224.0.0.102

  - v1MAC: 0000.0c07.ac##

    - ## = HSRP group

  - v2MAC: 0000.0c9f.f###

    - ### = HSRP group

#int g0/0
#standby [HSRP parameter]
#standby version 2
#standby 1 ip 172.16.0.254
  - VIP
#standby 1 priority 200
  - value•ip address determines active router
#standby 1 preempt
  - sets current router to active - when online - regardless of others in the topology

#show standby

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- VRRP: virtual router redundancy protocol

  - open standard

  - master/backup election

  - single master - per - subnet load balancing

  - Multicast: 224.0.0.18

  - vMAC: 0000.5e00.01##

    - ## = VRRP group

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- GLBP: gateway load balancing protocol

  - cisco proprietary

  - multi active - per - subnet load balancing

    - AVG: active virtual gateway, elected for the subnet

    - AVF: active virtual forwarders, up to four gateways within a single subnet
      **AVG can be an AVF

  - Multicast: 224.0.0.102

  - vMAC: 0007.b400.[##][##]

    - ##: GLBP group

    - ##: AVF

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Multilayer Switching

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- packet forwarding via layer 3 switching

- generally require a few commands to enable

#ip routing

  - enables packet forwarding

#no switchport

- disables layer 2 protocol/enables layer 3 on the port

#int vlan ####

- declares switched virtual interface(SVI) through which traffic is routed

#ip add ####
- address local to the subnet residing within the vlan
- address serves as the gateway for the vlan

**permits inter-vlan routing & packet forwarding w/o the ROAS topology

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STP

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- port role comprehension

- draw topology

- form connection matrix detailing connections in relation to interface per device

- determine root bridge

- bridge priority•bridge ID(MAC)

- WAVE I (designated ports + root ports)

**root bridge ports default to designated role

- cost•port #

- WAVE II (designated ports + alternate ports)

- bridge ID

**backup ports work in conjunction with designated ports connected to hubs

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- spanning-tree protocol

- anti broadcast-storming

- configures a singular path amongst interconnected switches, without loops

**automatic functionality(one of the fundamental behaviors of cisco switches)

- places interfaces(trunking ports) in 1 of 2 primary states(there are 4 in total)

- forwarding: sending and receiving frames

- blocking: receiving, but not sending BPDU messages in anticipation of a necessary state transition

**listening & learning are transition states from blocking to forwarding

**this is the redundancy measure spanning-tree provides

- assigns interface roles

- root port: primary forwarding port(lowest cost toward the root bridge)

- designated port: forwarding

- non-designated port: blocking

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#show spanning-tree

- displays root-bridge (switch with the lowest priority, MAC)

- displays current bridge information

**#show spanning-tree ?
- display more detailed information

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- PVST & PVST+

- per vlan spanning-tree

- allowing differing topologies per vlan

ex. 2 pvst instances with differing root bridges amongst the same devices

- PVST: ISL trunking protocol ONLY

- PVST+: adds 802.1q trunking compatibility

#vlan ####

**creates the aforementioned vlan, subsequently generating the spanning-tree instance

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#spanning-tree portfast

- portfast: skips listening/learning state

- not to be used on trunking ports

- allows immediate network access to end users

#spanning-tree portfast default

**enable portfast on all access ports

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#spanning-tree portfast trunk

- portfast(trunk)

**useful for server connections and router on a stick port

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#spanning-tree bpduguard enable

- BPDU guard: port shuts down up receipt of a BPDU message

- used in conjunction with portfast to avoid broadcast storms upon topology change

#spanning-tree portfast bpduguard default

**enables bpduguard on all portfast interfaces

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#spanning-tree bpdufilter enable

- disables BPDU messages on the port(essentially disabling STP)

#spanning-tree portfast bpduFilter default

**enables bpduFiltering on all portfast interfaces

- simply disable bpdu filtering on desired ports

**safer bet so as to avoid broadcast storms: bpduFilter is triggered

and port operates normally, subsequently triggering the bpduGuard and errDisabling

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#spanning-tree guard root

- root guard: stops interface from adopting new root bridge(regardless of priority/MAC)

- ensures topology behavior remains the same regardless of the priority of new switches

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#spanning-tree vlan 1 root primary

- sets current bridge to the root bridge for the aforementioned vlan

- useful for STP load balancing(differing port roles)

#spanning-tree vlan 1 root secondary

**sets backup root bridge

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IPv6

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- static route statements

- directly attatched(exit interface)

#ipv6 route 2001:db8:0:3::/64 s0/0
**inoperable configuration on ethernet interfaces

- recursive route(next-hop address)

#ipv6 route 2001:db8:0:3::/64 2001:db8:0:12::2

- fully specified(containing both exit interface and next-hop address)

#ipv6 route 2001:db8:0:3::/64 g0/0 2001:db8:0:12::2
**link local routes are always fully specified

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- static route types

- network

#ipv6 route 2001:db8:0:3:::/64 [next-hop]

- host

#ipv6 route 2001:db8:0:3::100/128 [next-hop]

- default

#ipv6 route ::/0 [next-hop]

- floating(recursive + administrative distance)

#ipv6 route 2001:db8:0:3::/64 2001:db8:0:12::2 2
**[2] > the default administrative distance of static routes(1)
**ensure administrative distance is greater than that of the current protocol in use

++
++

- SLAAC: stateless address auto configuration

- uses NDP to learn next-hop prefix and EUI64 to generate an IPv6 address on the interface

#ipv6 address autoconfig

++
++

- NDP: node discovery protocol

- replaces IPv4 ARP

- uses solicited node multicast address to learn neighboring MAC addresses

- NS/NA messages

- neighbor solicitation: what's your MAC?

- neighbor advertisement: my MAC is…

**messages form the MAC address table

RS/RA messages

- router solicitation: hello?
- send to FF02::2(all routers)

- router advertisement: i am…(node information)
- sent to FF02::1(all nodes)

**messages form the NDP table

++
++

- solicited node multicast address

- calculated from prefix + last 6 bits from configured unicast

ex. 2001:db8::1:f2a:4fff:fea3:b1

2001:db8:0000:0001:0f2a:4fff:fea3:00b1

a3:00b1

ff02::1:ffa3:b1

- interface nodes automatically join solicited node multicast address

++
++

- ipv6 extra utilities

- [::] unspecified ipv6
**0.0.0.0 equivalent

- [::1] loopback address

- ::/0 default route

++
++

- anycast

- one to any connection(first/nearest available)

- use any global unicast or unique local address

++

ex. #ipv6 address 2001:DB8:1:1::99/128 anycast

#show ipv6 int g0/0

++
++

- multicast ipv6

** there is no broadcast address in ipv6 techinally

- FF00::/8(FF00:: TO FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF)

- FF02::1 all hosts/nodes

- FF02::2 all routers

- FF02::5 all ospf

- FF02::6 all ospf DR/BDR

- FF02::9 all rip routers

- FF02::A all eigrp routers

**scope(distances/domains)

- FF01 interface scope(unpingable)

- FF02 link local scope(directly pingable; unroutable)

- FF05 site local scope(inter-LAN scope)

- FF08 organizational scope(WAN scope)

- FF0E global scope(publically routable[inter-WAN or internet])

++
++

- link local(protocol accomodations)

- auto generated
- interface ID is generated using EUI64 rules
- used for protocol parrings(LSAs, NDPs,etc)
**NDP: neighbor discovery protocol
- essentially ARP for ipv6
- used for next hop resolutions; DOES NOT ROUTE PACKETS

++

#ipv6 enable
**enables ipv6 and generated link local address

++
++

- Eui 64

- extended unique identifier
- "modified eui 64"
- converts 48bit mac to 64 bit interface identifier
- becomes the host portion of a /64 IPv6
- how to convert(done automatically on the interface)

ex. 1234 5678 90AB

divide mac in half

1234 56 || 78 90AB

insert 0xFFFE in the middle

1234 56FF FE78 90AB

invert the 7th bit

1[2]34 56FF FE78 90AB

0x2

0d2

0b0010

0b0000

0d0

0x0

1034 56FF FE78 90AB

++

782B CBAC 0867

782B CB || AC 0867

782B CBFF FEAC 0867

0x8

0d8

0b1000

0b1010

0d10

0xA

7A2B CBFF FEAC 0867

++

#show interfaces
*annotate the mac

#ipv6 address 2001:dB8::/64 eui-64
*configures the IPv6 address
- [global unicast identifier][EUI]

#show ipv6 interface brief
*compare address and mac relations

++
++

- unique local address(PRIVATE UNICAST)

  - FC::00/7(FC00:: to FDFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF)

  ex. FD45:93AC:8A8F:0001::/64

    - [FD] unique local identifier

    - [45:93AC] global ID
      **randomly generated to avoid overlaps

    - [0001] subnet ID

    - [::1] interface identifier

++
++

- global unicast address(PUBLIC UNICAST)

  - 2000::/3(2000:: to 3FFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF:FFFF)

  ex. 2001:DB8:8B00:0001::1/64

    - [2001:DB8:8B00] global routing prefix

    - [0001] subnet identifier

    - [::1] interface identifer

++
++

Find Network Address From Host

- 172.16.33.47/25

  convert address & subnet to binary

  10101100 00010000 00100001 00101111
  11111111 11111111 11111111 10000000

  multiply

  10101100 00010000 00100001 00000000

  **convert binary to decimal

  172.16.33.0/25

- 172.16.33.47/29

  10101100 00010000 00100001 00101111
  11111111 11111111 11111111 11111000

  10101100 00010000 00100001 00101000

  172.16.33.40
  .40 N
  .47 B
  .41 G
  .42 - .46 Hs

SHORTCUT

  **convert address to binary

  10101100 00010000 00100001 00101111

  **invert bits beyond subnet(network bits) to hosts(0s)

  10101100 00010000 00100001 00101000

  convert binary to decimal

  172.16.33.40/29

IPv6 Address Types

- global unicast address: host address

  - 2001:0DB8:8B00:1::1/64

    - 64 network bits

    - 64th bit(1) operates as the subnet identifier(last network bit)

    - 128th bit(1) operates as a host bit

    - global routing prefix: 2001:0DB8:8B00:1000/64
      **simply 0 host bits

  - 300D:00F2:0B34:2100::1200:0001/56

    - 56 network bits

    - 2[1]00 ~~~ 1 operates as the subnet identifier

    - global routing prefix: 300D:00F2:0B34:2100::/56

  - 2001:0DB8:8B00:0001:FB89:017B:0020:0011/93

    - 93 network bits

    - 017[B] ~~~ B CONTAINS THE BIT, that operates as the subnet identifier

      - B = 0xB

      - 0xB = 0d11

      - 0d11 = 0b1011

      - 0b[1]011 ~~~ 1 operates as the subnet identifier

      - 0b1011 becomes 0b1000

      - 0b1000 = 0d8

      - 0d8 = 0x8

      - 017[B] becomes 0178

    - global routing prefix: 2001:DB8:8B00:1:FB89:178::/93

IPv6 Address Simplification

- 2001:0DB8:000A:001B:20A1:0020:0080:34BD

  leading 0s

  - 2001:DB8:A:1B:20A1:20:80:34BD

- 2001:0DB8:0000:0000:0000:0000:0080:34BD

  consecutive 0s

  - 2001:0DB8::0080:34BD

  leading 0s

  - 2001:DB8::80:34BD

- 2001:0000:0000:0000:20A1:0000:0000:34BD

  consecutive 0s(can only be abbreviated once)

  - 2001::20A1:0000:0000:34BD

  leading 0s

  - 2001::20A1:0:0:34BD

GK

- 8 bits = 1 byte
- 1 byte = …2^3 2^2 2^1 2^0
- IPv4: 4 octets
  - 8 bit groups represented in decimal
  - 32 bits
    - /32 subnets
- IPv6: 8 quartets
  - 16 bit groups represented in hexadecimal
  - 128 bits
    - /128 subnets

Binary to Hexadecimal

- 0b11011011
  - 0b1101 0b1011
  - 0d13 0d11
  - 0xD 0xB
  - 0xDB
- 0b00101111
  - 0b0010 0b1111
  - 0d2 0d15
  - 0x2 0xF
  - 0x2F
- 0b10000001
  - 0b1000 0b0001
  - 0d8 0d1
  - 0x8 0x1
  - 0x81

Hexadecimal to Binary

- 0xEC
  - 0xE 0xC
  - 0d14 0d12
  - 0d1110 0d1100
  - 0d11101100
- 0x2B
  - 0x2 0xB
  - 0d2 0d11
  - 0b0010 0b1011
  - 0b00101011
- 0xD7
  - 0xD 0x7
  - 0d13 0d7
  - 0b1101 0b0111
  - 0b11010111

Multicast Addresses

- 224.0.0.1: all hosts(nodes) on a local network
- 224.0.0.2: all routers on a local network
- 224.0.1.1: NTP
- 224.0.0.5: ospf routers

Classful Addressing

- A: 1.0.0.0 - 126.255.255.255
  - /8
  **private: 10.0.0.1 - 10.255.255.255
- B: 128.0.0.0 - 191.255.255.255
  - /16
  **private: 172.16.0.0 - 172.32.255.255
- C: 192.0.0.0 - 223.255.255.255
  - /24
  **private: 192.168.0.0 - 192.168.255.255
- D: 224.0.0.0 - 239.255.255.255
  - /32
  - multicasting addresses
- E: 240.0.0.0 - 255.255.255.255
  - experimentakl use and testing

GK

- BLOS: beyond line of sight(hawkeye•other forms of satcom)

MPLS(Multi Protocol Label Switching)

- comparison to traditional switching(router function)
  - every router performs an IP lookup
  - traditionally, the router determines next-hop based on routing table
  - wasteful in terms of resources and time

- MPLS uses LABLES instead, to improve QoS, connectivity, and mesh topology capabilities
  - packets are tagged with MPLS labels
  - packets undergo "label-switching" as they travel through the internet, eliminating subsequent header examinations
    - **edge routers are the only two that investigate the header for the necessary information(source, destination, transport layer data, QoS, etc)

- HOW IT WORKS
  - assign labels to each route in the routing table(16-######)
    - ex. R1 - 1.1.1.1/32 == 100
      R2 - 2.2.2.2/32 == 200
  - share labels(LDP - label distribution protocol•automatic process)
  - build table-based forwarding table

++
++

- MPLS

  - show mpls neighbor

  - show mpls forwarding table'

  - encapsulation mpls

++

  #show mpls ldp neighbor

  #show mpls forwarding-table

  #interface g0/1
  #encapsulation mpls
  #mpls label protocol ldp
  #mpls ip

  #mpls ?

  # show mpls ldp bindings

  #show mpls ldp discovery

++
++

Etherchannel

- useful for load balancing between two devices(layers 2 & 3)

- directs 2 or more ports to act as one in the interest of increasing bandwidth
  **stp isolates frames/packets to a preferred route to avoid looping, rendering other links obsolete

- load is balanced based on flows
  **flows is the communication between two nodes(devices) in the network(consistent etherChannel path)

- calculations are made to determine etherChannel path
  - inputs for consideration
    - source MAC
    - destination MAC
    - source and destination MAC
    - source IP
    - destination IP
    - source and destination IP
    - layer 4 port(tcp/udp)

- interfaces(max of 8) must have identical configurations

- methods of etherChannel

  - PAgP(Port Aggregation Protocol)
    **cisco proprietary, dynamic etherchannel negotiation
  - LACP(Link Aggregation Control Protocol)
    **preferred, universal, dynamic etherchannel negotiation
  - Static etherChannel
    **manual etherchannel formation not requiring algorithm or discovery protocol

Security Policy

- manufactures default security configuration
**contains default passwords, configurations, and management information

Scripting

- issuing commands in similar fashion multiple times(batch/preset commands)
- powershell
  - speaks in cmdlets and cmds

++
++

Script 1

  - suspend output being written to the command prompt
  - calls the system command to produce a network configuration summary and redirects(pipes) its output to a text file
  - waits and prompts the user to press a key, then opens the report file in Notepad

++

  >@echo off
  >ipconfig /all > report.txt
  >echo Press a key to view report
  >pause > nul
  >notepad report.txt

++

  - reads the script from the parent folder to the command prompt

++

  >documents\test.cmd

++
++

Script 2

  - displays list of numerals

++

  >iHigh = 50
  >iLow = 10
  >for i = 1 to 5
    randomize
    iDisplay = int((iHigh - iLow + 1) * rnd + iLow)
    wscript.echo iDisplay
  >next

++

```
        >cscript documents\test.vbs

++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
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    Script 3

        - displays list of numerals & provides informative feedback

++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++

        >iHigh = 50
        >iLow = 10
        >iCount = 0
        >sTitle = "Number Count"
        >for i = 1 to 10
            randomize
            iDisplay = int((iHigh - iLow + 1) * rnd + iLow)
            wscript.echo iDisplay
            if iDisplay > 25 then
                iCount = iCount + 1
            end if
        >next
        >sMsg = cstr(iCount) + "numbers are greater than 25"
        >msgbox sMsg, vbOkOnly, sTitle
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++

    Script 4

        - displays system information

++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++

        >Get-WmiObject -Class Win32_OperatingSystem -ComputerName localhost

++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++

    Script 5

        - displays hostname

++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++

        >Get -WmiObject -Class Win32_OperatingSystem -ComputerName localhost | Select-Object -Property CSName

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    Script 6

        - displays all current running services

++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++

        >Get-Service

++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++

    Script 7

        - limits display of current running services to display name

++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++

    Script 8

        - limits lists of current services to display name and current state

++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++

        >Get-WmiObject win32_service | Select DisplayName, State | SortState, DisplayName

++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++s

    Script 9(Powershell: run as admin)

        - enable scripts without valid coding certificate

++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++

        >Set-ExecutionPolicy RemoteSigned

++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++



Switch GK

    - fundamental behaviors
        - learn
            - adds MAC address table entries
        - flood
            - sends frame out of all ports(multicast frames) ACCEPT receiving port
            **switch also floods when unicast frame is received from MAC address foreign to the table
        - forward
            - forwards frame to port that responds with the destination Mac

    - interface(port) security

        - allowed on trunk/access ports
            **not allowed on dynamic auto/desirable ports

        - errdisable recovery
            - re-enables interfaces, after a specified time, if enabled for the listed
              cause of disablement

        - violation modes
            - shutdown
                - err-disabled state(shutdown)
                - generates syslog and/or SNMP message
                - violation counter + 1
            - restrict
                - discards traffic from unauthorized mack
                - interface remains up
                - generates syslog and/or SNMP message
                - violation counter + 1 / per frame
            - protect
                - discards traffic from unauthorized mack
                - interface remains up
                - DOES NOT generate syslog and/or SNMP message
                - DOES NOT increment the violation counter

        - MAC address aging
                - default: 0min (no age out)
                    - static addresses won't age out
                    **sticky MACS don't age out
                    **#switchport port-security aging static
                        - enables aging for static addresses
                - type
                    - absolute
                        - address is learned, aging timer starts, and is
                          removed once timer expires
                    - inactivity
```

- address is learned, aging timer starts, and
restarts upon every frame reception

Hawkeye Transec Terms

- CSP: certificate signature request
- RSP: remote security package

GK

- adding IPs to management laptop “advanced” tab sets management laptop as a host on remote device network

Hawkeye Manual Point

- go to working hawkeye
- iPhone measurement app
- place phone(long side) behind left stand
- record the displayed level
- place phone(short side) at the L3 line
- record the displayed compass angle
- stow broken hawkeye
- adjust peak and poll using controller for highest signal strength

Load Bootfile Hawkeye

++
++

- annotate boot file(for SES)
- find and copy contents of bootfile
  *cli bootfile contains “echo” commands
- putty into the modem
- login

  #lway stop
- paste contents

  #lway start

  #ttelnet

  #rtp
  **displays neighbors

++
++

GK

- OEM: original equipment manufacturer software
  - single-use license; tied to the original machine

Integrated Development Environment

- programming tool, designed to make working with a given language easier
  ex. NetBeans - Java

Interpreted Programming Languages(compiled in realtime by an interpreter)

**scripting languages

- JavaScript
- Perl
- Python

Compiled Programming Languages(converts to machine code when compiled)

- C++
- C#
- COBOL
- PASCAL

Data Types

- integer: 1 12 546 15 5
- floating-point: 4.5
- boolean: true(1) OR false(0)
- characters: a D T @ $
- strings:
  - “XYZ” \hello world\
  ** “hello \world\ ”
  **contains double quotations
- unicode: data representation format that can encode the widest range of characters

Servers

GK

- ports describe virtual connections to specific services
  **ips describe physical connections(locations)
- port types:
  - tcp: Transmission Control Protocol(connection based port)

- udp: user datagram protocol(connectionless)

- netstat
  - windows command
  - displays current virtual connections

Creating and Using a Database

- databases organize stored data in a manner that makes it relatively easy to retrieve data
  - Apache Open Office
  **SQL underlying equivalent of CLI

Database syntax

- “
  - “monty python”

  - specify a match to the exact phrase as typed

- +

  - +snake +Python

  - words that must be found in the documents exactly as typed

- -

  - python -Monty

  - excluded the following word

- OR

  ** I(pipe)

  - snake I python

  - find either word

- *

  - genius * python

  - represents unknown words in between the known

Windows prompts(see link above)

- “XXXX” /?
  ***XXXX” - - help
  - generates options for given command

- ipconfig
  - generates PC networking information

- ipconfig /all
  - generates detailed networking devices information

- ctrl + shift + esc
  - launch tack manager
  **right click task bar
  **right click start button
  **windows + x

- taskkill
  - “end task”

- ctrl + f5
  - reloading page with renewed cache

- right click “this pc” in file explorer

  - launch Computer Management Console

- diskmgmt.msc

  - launches disk management utility
  **windows + x

- netstat
  - displays current protocol(4), local address, foreign address, and link state of current internet connections
  **netstat -r
    - host routing table
  **netstat -an
    - current TCP connections

CISCO CLI capabilities(search “Command Reference” on cisco.com)

++
++

- mtu adjust

  - adjust packet size per transmission

++

  #interface g0/0/0
  #ip mtu #######

++
++

++
++

- mss adjust

  - adjust TCP segment size per payload

++

  #interface g0/0
  #ip tcp adjust-mss #######

++
++

++
++

- ospf passive interface

  - discontinues LSAs

++

  #router ospf 1

#passive-interface default
**LSAs will not be sent from interfaces participating in “ospf 1”
#no passive-interface g0/0
**LSAs will be sent explicitly from aforementioned interfaces

++
++

++
++

- ospf cost

- adjust cost in relation to reference bandwidth(propagate throughout the domain)

++

#router ospf 1
#auto-cost reference-bandwidth #######

++
++

++
++

- ip phone

- configure switch interfaces(voice and data vlan)

- configure router-on-a-stick

SW1#interface g0/1
#switchport mode access
#switchport access vlan 10
#switchport voice vlan 20
#no shutdown
#exit

#interface g0/10
#switchport mode trunk
#switchport mode allowed vlan 10,20
#no shutdown
#exit

RT1#interface g0/10
#no shut

#interface g0/10.10
#encapsulation dot1q 10
#no shutdown
#exit

#interface g0/10.20
#encapsulation dot1q 20
#no shutdown
#exit

++
++

++
++

- basic layer 2 etherchannel

#int range g0/0-2

#switchport mode trunk

#channel-group ### mode [mode parameter]

#port-channel load-balance [load-balance parameter]

++

- basic layer 3 etherchannel

#int range g0/0-2

#no switchport

#channel-group ### mode [mode parameter]

#port-channel load-balance [load-balance parameter]

#int po###

#ip add [address][subnet]

#ip route 0.0.0.0 0.0.0.0 [next hop etherchannel interface ip]

++

- etherChannel load-balance considerations

#show etherchannel load-balance

- etherchannel summary

#show etherchannel summary

#show etherchannel port-channel

#show

++
++

++
++
- show cdp neighbours

- displays directly connected hostname + interface
**may/may not disclose all

#show cdp neighbors

**show cdp ?

**cdp run
- enables cdp

**cdp enable
- enables cdp discovery

++
++

Virtual Route Forwarding(VRF)

++

- create vrf instances
  #ip vrf customer1
  #ip vrf customer2
  #show ip vrf

- enable vrf per interface
  #interface g0/0
  #ip vrf forwarding customer1
  #ip address 192.168.0.1 255.255.255.252
  #no shutdown

  #interface g0/1
  #ip vrf forwarding customer2
  #ip address 192.168.0.1 255.255.255.252
  #no shutdown

- display vrf instances
  #show vrf

- display interfaces participating in vrf routing
  #show ip vrf interfaces

- create vrf routes(address families)
  #router ospf 1
  #address-family ipv4 vrf customer1
  #network 0.0.0.0 255.255.255.255 area 0

  #router ospf 2
  #address-family ipv4 vrf customer2
  #network 0.0.0.0 255.255.255.255 area 0

- display vrf routing tables
  #show ip route vrf customer1
  #show ip route vrf customer2

- ping across a vrf
  #ping vrf customer1 192.168.0.1
  #ping vrf customer2 192.168.0.1

++

- route distinguisher(RD): virtual tag used to identify vrf association
  ex. VRF Customer1
    RD: 65000:10

    VRF Customer2
    RD: 65000:20

- route tag(RT): virtual tag used to identify ingress/egress targets inside/outside the vrf
  ex. VRF Customer1
    RD: 65000:10
      RT export: 65000:100
      RT import: 65000:100

   VRF Customer2
    RD: 65000:20
      RT export: 65000:200
      RT import: 65000:200
  **sharing RTs allows inter-VRF communication

++
++

- port security
  - enable port security
  - static Mac
  - assign violation mode
  - enable static address aging
  - assign aging time(minutes)

  #interface f0/1
  #switchport port-security
  #switchport port-security mac-address 000a.000a.000a
  #switchport port-security violation restrict
  #switchport port-security aging static
  #switchport port-security aging time 10

  #show port-security interface f0/1

++

  #interface f0/2
  #switchport port-security
  #switchport port-security mac-address 000a.000a.000a
  #switchport port-security violation shutdown
  #switchport port-security aging static
  #switchport port-security aging time 10

  #show port-security interface f0/2

++

  #interface f0/3
  #switchport port-security
  #switchport port-security mac-address 000a.000a.000a
  #switchport port-security violation protect
  #switchport port-security aging static
  #switchport port-security aging time 10

  #show port-security interface f0/3

++
++

- mGRE & DMVPN
  - dynamic multipoint virtual private network

  - NHRP: next hop resolution protocol

    - NHS: next hop server(hub/hq router)

    - NHC: next hop client(spoke)

  - NHRP configuration

    - R1(NHS)
      - WAN 1.1.1.1
      - tunnel 192.168.10.1/24

    - R2(NHC)
      - WAN 2.2.2.1
      - tunnel 192.168.10.2/24

    ***Internet
      - NHS 1.1.1.2
      - NHC 2.2.2.2
      - NHC 3.3.3.2

++

- NHRP CLI

  - GRE: generic routing encapsulation
    - nhs = next hop server

    - nhc = next hop client

++

- Static routing(more outers = more map entries)

```
R1# interface tunnel1
R1# ip address 192.168.10.1 255.255.255.0
R1# tunnel source f0/0
R1# tunnel mode gre multipoint
R1# ip nhrp network-id 1
R1# ip nhrp map 192.168.10.2 2.2.2.2

R2# interface tunnel1
R2# ip address 192.168.10.2 255.255.255.0
R2# tunnel source f0/0
R1# tunnel mode gre multipoint
R2# ip nhrp network-id 1
R2# ip nhrp map 192.168.10.1 1.1.1.1

show dmvpn

ping x.x.x.x source x.x.x.x

show run | sec
```

++

- OSPF over DMVPN(Phase I - hub + spoke(s) data flows)

```
R1# interface tunnel1
R1# ip address 192.168.10.1 255.255.255.0
R1# tunnel source f0/0
R1# tunnel mode gre multipoint
R1# ip nhrp network-id 1
R1# ip nhrp map multicast dynamic

R2# interface tunnel1
R2# ip address 192.168.20.1 255.255.255.0
R2# tunnel source f0/0
R2# tunnel mode gre multipoint
R2# ip nhrp network-id 1
R2# ip nhs 192.168.10.1
R2# ip nhrp map 192.168.10.1 1.1.1.1
R2# ip nhrp map multicast 1.1.1.1
```

++

- OSPF over DMVPN(Phase II(tweaking protocol) - spoke + spoke(s) data flows)

```
R1# interface tunnel1
R1# ip address 192.168.10.1 255.255.255.0
R1# tunnel source f0/0
R1# tunnel mode gre multipoint
R1# ip nhrp network-id 1
R1# ip nhrp map multicast dynamic
R1# NO IP NEXT-HOP-SELF OSPF 1

R2# interface tunnel1
R2# ip address 192.168.20.1 255.255.255.0
R2# tunnel source f0/0
R2# tunnel mode gre multipoint
R2# ip nhrp network-id 1
R2# ip nhs 192.168.10.1
R2# ip nhrp map 192.168.10.1 1.1.1.1
R2# ip nhrp map multicast 1.1.1.1
```

++

- OSPF over DMVPN(Phase III(tweaking nhrp) - spoke + spoke(dual hub) data flows)

```
R1# interface tunnel1
R1# ip address 192.168.10.1 255.255.255.0
R1# tunnel source f0/0
R1# tunnel mode gre multipoint
R1# ip nhrp network-id 1
R1# ip nhrp map 192.168.10.3 3.3.3.3
R1# ip nhrp map multicast 192.168.10.3
R1# ip nhrp map multicast dynamic
R1# IP NHRP REDIRECT

R3# interface tunnel1
R3# ip address 192.168.10.3 255.255.255.0
R3# tunnel source f0/0
R3# tunnel mode gre multipoint
R3# ip nhrp network-id 1
R3# ip nhrp map 192.168.10.1 1.1.1.1
R3# ip nhrp map multicast 192.168.10.1
R3# ip nhrp map multicast dynamic
R3# IP NHRP REDIRECT

R2# interface tunnel1
R2# ip address 192.168.20.1 255.255.255.0
R2# tunnel source f0/0
R2# tunnel mode gre multipoint
R2# ip nhrp network-id 1
R2# ip nhs 192.168.10.1
R2# ip nhs 192.168.10.3
R2# ip nhrp map 192.168.10.1 1.1.1.1
R2# ip nhrp map 192.168.10.3 3.3.3.3
R2# ip nhrp map multicast 1.1.1.1
R2# ip nhrp map multicast 3.3.3.3
R3# IP NHRP SHORTCUT
```

++
++

- IKEv1(6)

```
**local and peer tunnel settings are required to match
- acl
- isakmp policy(phase 1)
        - isakmp key exchange
- ipsec transform-set (phase 2)
        - encrypting the tunnel
- crypto map
        - vpn instructions
- apply crypto map
        - outside interface
- show crypto ipsec security associations

R1#show version
**verify presence of "security license"

R1#license boot module c1900 technology-package securityk9

R1#access-list 100 permit ip 192.168.1.0 0.0.0.255 192.168.3.0 0.0.0.255
! ip <local LAN> <wildcard> <peer LAN> <wildcard>

R1#crypto isakmp policy 10
R1#encryption aes 256
R1#authentication pre-share
R1#group 5
R1#exit

R1#crypto isakmp key secret key address 209.165.200.1
! key [secret] <pre-shared key>
! address <peer wan link>

R1#crypto ipsec transform-set R1->R3 esp-aes 256 esp-sha-hmac
R1#mode tunnel
R1#exit

R1#crypto map IPSEC-MAP  10 ipsec-isakmp
R1#set peer 209.165.200.1
R1#set pfs group5
```

R1#set security-association lifetime seconds 86400
R1#set transform-set R1->R3
R1#match address 100
R1#exit

R1# interface g0/0
R1#crypto map IPSEC-MAP

show crypto ipsec sa
show crypto isakmp sa
show crypto session

++
++

- access list(3)
  - ***standard -> 1-99 - - applied closest the recipient
    - ***denies or permits source IP address
  - ***extended -> 100-199 - - applied closest to the sender
    - *** denies or permits source IP address
    - *** denies or permits destination IP address
    - *** denies or permits port(service)
  - ***access lists are accompanied with an IMPLICIT DENY ALL statement at the end
  - end acl statement with "permit any"
  - ***rule order dependent(prioritize by specificity)
  - access-list configuration
  - access-list application
  - test

++

R1#ip access-list standard 1
R1#deny 192.168.1.101 0.0.0.0
R1#permit any

R1#interface g0/0
R1#ip access-group 1 inside

R1#interface g0/1
R1#ip access-group 1 outside

ping 192.168.1.101
ping 192.168.1.102

++

R1#ip access-list extended 100
R1#deny tcp 192.168.1.101 0.0.0.0 eq 80 200.0.16.254 0.0.0.0
R1#permit ip any any

R1#interface g0/0
R1#ip access-group 100 inside
R1#exit

R1#interface g0/1
R1#ip access-group 100 outside

ping 192.168.2.100
ping 192.168.2.101
show traceroute
  - ***extended traceroute: allows trace conditions including port #(transport layer - UDP/TCP connections)

++
++

- NAT(network address translation)

  - static NAT(4)[source nat translation]
    - static nat statement (inside local{private ip} - inside global{public ip})
    - inside interface
    - outside interface
    - show ip nat translations

    natROUTER#ip nat inside source static 192.168.20.10 200.100.50.1

    natROUTER#interface g0/0
    natROUTER#ip nat inside
    natROUTER#exit

    natROUTER#interface g0/1
    natROUTER#ip nat outside
    natROUTER# exit

    show ip nat translations
    show ip nat statistics

++

  - dynamic translation(NAT POOL)(5)
    - inside interface
    - outside interface
    - acl
    - NAT pool
    - dynamic NAT statement(dNAT OR PAT)

    natPOOL#interface g0/0
    natPOOL#ip nat outside
    natPOOL#exit

    natPOOL# interface g0/1
    natPOOL#ip nat inside
    natPOOL#exit

    natPOOL#ip access-list standard 1
    natPOOL#permit 192.168.1.0 0.0.0.255
    natPOOL#deny any

    natPOOL#ip nat pool MYPOOL 200.100.50.2 200.100.50.254 netmask 255.255.255.0
    ***translate inside local list to inside global list

    OR

    natPOOL#ip nat inside source list 1 interface g0/0 overload
    ***translates inside local to outside interface address
    ***overload: assigns each source a port # for translation exclusivity

    show ip nat translations
    show ip nat statistics

++
++

- loopback(3)
  - create loopback
  - enable in routing protocol
  - verify connectivity

  interface loopback0
  ip address 192.168.1.1 255.255.255.255
  exit

  router ospf 1
  network 192.168.1.1 0.0.0.0 area 0
  exit

  show ip interface brief
  show ip route
  ping 192.168.1.1
  traceroute 192.168.1.1

++
++

- tunneling(3)
  - configure interfaces
  - configure router routes
  - configure tunnel
  - input static ip routes

  R1 interface f0/1
  R1 ip address 192.168.1.1 255.255.255.252
  R1 no shutdown
  R1 exit

  R3 interface f0/1
  R3 ip address 192.168.1.2 255.255.255.252
  R3 no shutdown
  R3 exit
  R3 interface f0/2
  R3 ip address 192.168.1.6 255.255.255.252
  R3 no shutdown
  R3 exit

  R2 interface f0/2
  R2 ip address 192.168.1.5 255.255.255.252
  R2 no shutdown
  R2 exit

  R1 ip route 192.168.1.4 255.255.255.252 192.168.1.2

  R3 ip route 192.168.1.0 255.255.255.252 192.168.1.6

  R1 Tunnel0
  R1 Ip address 192.168.1.13 255.255.255.252
  R1 tunnel source f01/1
  R1 tunnel destination 192.168.1.6 255.255.255.252
  R1 tunnel mode gre ip
  R1 exit

  R2 tunnel0
  R2 ip address 192.168.1.14 255.255.255.252
  R2 tunnel source f0/2
  R2 tunnel destination 192.168.1.2 255.255.255.252
  R2 tunnel mode gre ip
  R2 exit

  R1 ip route 192.168.1.16 255.255.255.252 192.168.1.10

  R2 ip route 192.168.1.12 255.255.255.252 192.168.1.9

++
++

- serial interface(requires layer 3 encapsulation[4])
  - interface
  - clear encapsulation
  - enter encapsulation
  - verify

  interface s0/1/0
  no encapsulation
  encapsulation ppp
  exit

  show ip interface brief

++
++

- dhcp server(4)
  - interface configuration
  - hostname
  - dhcp excluded address(s)[nets/broads/gates]
  - ip dhcp pool(s)

  interface g0/0
  ip address 192.168.0.1 255.255.255.0
  no shutdown
  exit

  hostname R1

  service dhcp

  ip dhcp excluded-address 192.168.0.1
  Ip dhcp excluded-address 192.168.0.0
  Ip dhcp excluded-address 192.168.0.255

  ip dhcp pool 192.168.0.1
  network 192.168.0.0 255.255.255.0
  default-router 192.168.0.1
  dns-server 8.8.8.8
  exit

  show ip dhcp binding

++

- dhcp relay
  - interface config mode(client side)
  - assign ip helper address(dhcp server address)

  **dhcp server ip: 192.168.0.254

  interface g0/0
  ip helper-address 192.168.0.254

++

- dhcp client
  - interface config mode(server side)
  - dhcp client declaration

  interface g0/0
  ip address dhcp
    **ip dhcp client lease ###
      - custom lease

  show dhcp lease
  show dhcp server

++
++

- interfaces
  - access & trunks

- vlans

- transport
  - telnet

++
++

- ssh configuration(9)
  - interface configuration
  - hostname
  - enable
  - domain name
  - username password
  - crypto keys

- line vty
- enable ssh
- test ssh

interface g0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
exit

hostname R1

enable password password

ip domain-name domain-name.com

username username password password

crypto key generate rsa
1024

line vty 0 15
local login
transport input ssh
exit

ip ssh version 2

c:\> ssh -l username 192.168.1.1

++
++

- routing protocols
  - static
  - rip
    - ver1 & 2

++
++

- ospf(7)
  - hostname
  - interface
  - router protocol configuration
  - show ip routes(O)
  - show connected ospf routers
  - show ospf

hostname hostname

interface f0/0

ip address 192.168.1.1 255.255.255.0
no shutdown
exit

router ospf 1
router-id #.#.#.#
network 192.168.1.0 0.0.0.255 area 0
exit

show ip protocols

show ip route ospf

show ip ospf neighbor

show ip ospf database

show ip ospf interface brief

++
++

- password
  - standard & encrypted

- banner

- subinterfaces

- ip domain - name

- hostname

- line vty

++
++

GK

- hostname: unique name for a device on the network

- ip - domain name
  - unique name for a network on the internet
  - associated with a particular organization

- DNS: domain name system
  - coverts domain names into IPs
  - primarily used to contact the correct network for services

Subnetting

- subnet grid - - - - - - - - wildcard grid
  - /32 0 IPs - - - - - - - - 0 IPs
  - /31 2 IPs - - - - - - - - 1 IP
  - /30 4 IPs - - - - - - - - 3 IPs
  - /29 8 IPs - - - - - - - - 7 IPs
  - etc

- subnet grid II - - - - - - - - wildcard grid
  - /32 255.255.255.255 - - - - - - - - 0.0.0.0
  - /31 255.255.255.254 - - - - - - - - 0.0.0.1
  - /30 255.255.255.252 - - - - - - - - 0.0.0.3
  - /29 255.255.255.248 - - - - - - - - 0.0.0.7
  - /28 255.255.255.240 - - - - - - - - 0.0.0.15
  - /27 255.255.255.224 - - - - - - - - 0.0.0.31
  - /26 255.255.255.192 - - - - - - - - 0.0.0.63
  - /25 255.255.255.128 - - - - - - - - 0.0.0.127
  - /24 255.255.255.0 - - - - - - - - 0.0.0.255
  - etc

- subnet concept I
  - /0 contains 1 /0 network
  - /0 contains 2 /1 networks
  - /0 contains 4 /2 networks
  - /1 contains 1 /1 network
  - /1 contains 2 /2 networks
  - /1 contains 4 /3 networks
  - etc

- subnet concept II
  - every network reserves 3 IPs(network ID - gateway - broadcast) regardless of structure
  **/30(4 IPs :: 1 host) - acceptable configuration: point to point(router to router) usage of the gateway and host IPs

- subnet concept

- /24: 11111111.11111111.11111111.00000000

- 0s indicate host bits/changeable bits

- octets with eight, 1s indicate the preservation of the IPs digits

ex. 192.168.1.0/24
- 192.168.1.0
- 192.168.1.1
- 192.168.1.3
- etc

- subnet concept

- objective: divide "given network" into /24 networks
"given subnet" -> A.B.C.D
"/24 -> 255.255.255.0
"/24 = 11111111.11111111.11111111.00000000

- the last 1 indicates the interval of the network
**count from 0, right to left
- interval = 256 IPs per network

ex. 192.168.1.0/24
- network: .0
- gate: .1
- broadcast: .255
- next network: 192.168.2.0/24

**subnetting rule

- boundary intervals are equal and always begin on .0
**the math may check out according to vlsm, but essentially there are fixed ip address ranges

- 172.16.0.4/29

**gateway 172.16.0.10 is invalid

- /29 = 3 host 0s = 8

- valid increments 172.16.0.0/29 + 8 -> 172.16.0.8 etc
**172.16.0.4 and 172.16.0.10 are located in separate subnets

**adjacent vlsm must adhere to increment boundaries

ex. [172.16.0.0/30](active) [172.16.0.4/30](unoccupied)

[172.16.0.0/29](occupied) [172.16.0.8/29](FIRST USABLE /29 BLOCK)

- Network Building

- device schematic

- list IP types(hosts, physical, virtual, etc)

- tally IPs per type

- list subnets(consider any form of separation)

- tally IPs per subnet(2 hosts, 45 hosts, etc)

- ip totals/subnet(IDs - gates - broads included)

- LDIF/ipMatrix
**vlsm - largest to smallest network
**consider classful ranges, and vlsm increments

- update schematic

GK

- RJ45 build(T5688)(OW - O - GW - Bl - BlW - G - BrW - Br)

Taclane Configuration(rebuild peer enclave)

- primary objective: rebuilding plain(red) & cypher(black) text in taclane
**red & black IPs
- neighborship exchange w/ RHN

- verify cik key serial and taclane serial
**do not power on w/o cik key NOR remove cik during operation

- kg250 console cable

- kg250 console application

- management -> pt

- network config menu

- denote red default gateway

- configure management IP
**ip will be the gateway

- denote ipv4 address
**enter address into Firefox(https://#.#.#.#)

- login

- open putty via desktop shortcut

- open LDIF(located in team packet)
- pt is in sipr
- ct is in xless

- putty(taclane config)
- user & pass
- snmp pass
- pt gate
- ct gate
- ping configured ip(verification)

- open taclane in GUI

- provision cik

- open “kg250x vers 2.iwks”

- skl
- radio EAD
- transec EAD
- open keys
- select unused key
- load
- send wakeup
- open automated SI keys
- verify firefly key consistency(gui)

- build remote route to RHN
- provided in the ASA

3 Phases of Care

- care under fire
  - return fire
  - command “return fire”
  - command “move to cover”
  - command “perform self aid”
  - secure fire superiority
  - post security by casualty
  - life saving aid
    - hasty tourniquet(as close to the heart as possible)
  - move casualty to cover
- tactical field care
  - MARCH PAWS algorithm
    - massive hemorrhaging
      - re-evaluate tourniquet
      - visual bloodsweep
      - deliberate tourniquet(never surround the wound with tourniquets)
      - check for junctional wounds(neck, armpits, groin)
      - extremities bloodsweep
    - airway
      - head tilt, chin lift
      - jaw thrust
      - nasopharyngeal airway(right nostril first)
    - respiration
      - check for equal rise+fall of the chest(ear to belt buckle)
      - find wound
        - remove top
        - rake(outward, top to bottom)
      - prep chest seal
        - cover wound with back of hand
        - exhales(3x)
        - chest seal(place seal at bottom of the exhale)
      - clean wound
      - place yourself opposite the wound, beside the casualty
        - use momentum to roll casualty onto your knees
      - repeat care for exit, back wounds affecting respiration
      - burp wound
      - decompression needle
        - find nipple line(find areas just below, on the casualty’s side
        - inspect the needle
        - insert into previously identified area(5th intercostal space[between the ribs])
        - remove needle, leaving tube inserted
        - secure would with tape
- tactical medivac
  - 9 line
    1 location
    2 callsign/suffix/frequency
    3 # of patients by precedents
    4 special equipment needed
    5 # of patients by type
    6 method of marking pickup sight
    7 security of pickup sight
    8 patient nationality
    9 terrain description/ NBC contamination
    ***radio etiquette is essential
      - numbers(cero, tree, fower, fife, ait, niner)
      - military alphabet
      - break, line, over, copy, say again, out
    ***triage
      - massive casualty procedures
      - prioritising supplies and assignment

skl(1523 comsec)
- ensure sincgars is on/no comsec loaded/ FH/ knob on LD
- eqs
- RT1523
- plats
- FCTX ground
- FCTX 1523
- load
- ok
- next
- connect SKL -> add/fill
- send
- load
- stow
- 1
- follow prompt

1523 configuration
- insert battery
- secure cover(groove)
- knob(TST)
  - done?
- skl -> aud/fill
- knob(LD)
- press menu(comsec to CT)
- press menu(mode to FH)
- press clear
- sto(stores)
- press time
- enter Julian date(knuckle trick)
- store
- Zulu time(world clock)
  - hours+min(1130)
    **current time(11:29:59)
  - min+sec(305_)
    - oscillate the last digit(the approaching second)
    - sto to sync radio time

Cisco Commands

Switch commands

- show int g#/# switchport - displays name, status, admin mode, operational mode
- show port-security
- show port-security interface g#/# - interface security info
- show errdisable recovery
- switchport port-security mac-address sticky - converts all dynamically learned MAC addresses to sticky MACS
- vlan database - enters vlan database configuration mode
- vlan 59 name Data - creates vlan 59 and names it “Data”. This is added to the vlan.dat file on the flash.
- interface range f0/1-44 - enters configuration mode for the entire range of interfaces
- switchport mode access - sets the port as an access (user) port
- switch port access vlan 59 - assigns the port to a vlan 59
- switchport mode trunk - sets a port to trunk vlans(do not use spanning-tree pordtfast on a trunk)
- show vlan - displays vlan information
- show vlan id 59 - displays vlan information about only vlan 59
- show interface vlan 59 - displays interface characteristics for the specified vlan

- show Mac address-table - displays the current MAC address forwarding table

- delete flash:vlan.dat - removes the vlan database from flash memory

- spanning-tree portfast - takes the port immediately to the STP fForwarding state. Only used for switchport mode access ports and end devices(IP phones and computer hosts)

- mdix auto - allows straight or crossover cables to be used on a port using auto pinout negotiation

Xless commands

- service-module Gi1/0 session - allows console access to xt2s module from xt2r router

- Ctrl Shift 6, then X - allows return from xt2s module to xt2r router session

- Enter - allows return to xt2s session from xt2r if the service-module session was already active

- show ospfv3 neighbor - shows the ospfv3 neighbor table(xless+sipr)

- show ip ospfv3 neighbor - shows the ospfv3 neighbor table(nipr)

- sh ospfv3 data - shows the ospfv3 data

- sh ip pin neighbor - display all pin neighbors the router discovered

- sh ip route - displays the contents of the ip multicast routing table. Used for viewing taclane multicast

- sh arp - show the routes over the vlans and over vlans 501 and 101

Router commands

- enable(en) - enter or enable Privileged Exec mode
  ** "router#" - you are in privileged exec mode

- config t - used to enter configuration mode. Allows changes to the device
  ** "router(config)#" - you are in configuration mode

- interface f0/0 - used to enter interface configuration mode to make changes on an interface
  ** "router(config-if)# - you are in interface configuration mode

- exit - takes you out of configuration for that mode into the previous mode
  ** Ctrl Z - keystroke for exit

- tab - auto completes a command

- ? - provides a help list of available commands

- ping - used to test network connectivity to a distant ip address

- trace - used to trace the path in a network to a distant ip address

- Ctrl Shift 6 - stops the trace test and other activities

- do - allows you to do privileged exec commands in the configuration mode

- show run - displays the running configuration running in RAM

- show run | b router ospf 21 - show running configuration beginning from router ospf

- show run int g0/48 - shows running configuration for interface g0/48

- show run int g0/0.358 - shows running configuration for sub-interface g0/0.358

**(config-subif)::encapsulation ?
**(config-subif)::encapsulation dot1q ?
**(config-subif)::encapsulation dot1q vlan 1
***(?) or (*prompt* ?) lists available commands

- show start - shows startup configuration on the NVRAM

- show flash - show files in the flash

- show version - show iOS version and configuration registrer value

- show interface g0/0 - show statistics for a specific interface (such as g0/0)

- show ip interface brief - show a summary of all interfaces including status and ip address assigned

- show int description - show status of vlan and other interfaces

- show arp - shows the Address Resolution Protocol (ARP) table

- show ip route - shows the routing table currently being used

- show ip ospf - shows basic information about the OSPF routing process

- show ip ospf interface - shows basic information as it relates to all interfaces

show ospf neighbor

show ip ospf neighbor - lists all ospf neighbors and their states

- show ip ospf neighbor detail - displays a detailed list of OSPF neighbors

- show ip ospf database - displays the contents of the ospf database

- term mon - sends debugging messages to the chosen terminal line

- debug ip ospf events - displays all ospf events as they occur in real time

- clear ip ospf process - resets the ospf process, forcing ospf to resend hellos and recreates neighbors, database, and routing table

- undebug all - stops the debugging process

- sh ip nhrp nhs - displays the next hop servers for next hop server resolution. Show if tunnels are responding. RE = responding. E = expecting a reply.

- copy run start - saves the running configuration to the start up configuration and NVRAM

- copy flash start - copies the specified file to the start up configuration and NVRAM

- copy tftp flash - copies the specified file to the flash via tftp server specified

- erase start - used to erase the saved configuration.
  ** "write erase"

- delete flash:old_running.cfg - deletes a specified file from the flash. Never do delete flash by itself. This will erase the iOS.

- more flash:JNN-64130-NFWHS - shows the contents of a specified file on the flash

- config-reg 0x - changes the configuration register on a device

- reloads - reboots the device

- logging sync - when in line config mode, turns on automatic redisplay of command line interrupted by console messages

RJ45(ethernet) cable build
- cut
- strip cover
- fan cables
- insert exposed cables to RJ45 bit
- align the colours
- crimp

Dispatch Book
- Operator 5984-E(license)
- TC 5984-E(license)
- Two copies SF 91(Operator report of motor vehicle accident)
- Two copies of DD 518(Accident ID card)
- III Roadside Inspection
- DA 5988(PMCS) or DA2404
- QA/QC inspection checklist
- DA Form 5987-E(commander or designated rep if less than 7 days
- DA Form 5987-E(BN commander or designated rep if greater than or equal to 8 days)

BII
- chock block
- drip pan
- highway warning kit
- serviceable fire extinguisher
- unexpired first aid kit
- serviceable safety strap
- trailer latched with lights/chains/safety pin
- fire point gear
  - pick axe
  - shovel
  - axe
  - fire extinguisher

90F Radios(SINCGARS)
- Component List
  - **BOM: acronym for component list
  - mount
  - transmitter
    - E/F
    - **RT-1523x
  - amplifier
    - **92F amps carry '5158' serials(last 4 digits)
  - bracket, mounting
    - 2 parts
  - cable assembly(radio frequency)
  - amplifier(frequency multiplier)
  - amplifier adapter(mount extension)
  - cable assembly(special purpose, electric)
- configuration
  - secure transmitter to radio mount
  - secure microphone to 'RXMT'
  - secure antenna cable to top left receptacle on transmitter
  - mode
    - single channel plain text(unsecure)
    - **time sync is required for secure comms
  - secure antenna cable to transmitter
  - place selector on 'ld(load)'
  - menu
    - vol(9)
    - ch.1(matching paired transmitter)
    - pwr(low)
      - distance dependent
    - paired frequency
    - mode(sc[single channel])
    - comsec(pt[plain text])
  - **sq(squelch) on: regular use
  - **stby: battery saving mode(transmitter inoperable)

switching & routing
- concepts
  - ***CONNECTION ENDS MUST MATCH(for the most part)
    - ports, ospf, call routes, etc
  - prioritize colorless
    - transmission enclave
  - trunk ports: receptacle -> receptacle
    - utilized for network interconnect
    - ***carries multiple VLANs
  - access ports: receptacle -> end device/user
    - ***associated w/ ONE VLAN
  - ospf: open shortest path first
    - tunnel between networks
      - non passive: continuous ping(essentially)
  - mtu
    - maximum transmission/largest packet size
  - snmp
    - network monitor/enclave map
  - dhcp
    - automatic search and acquisition of first available ip address
- commands
  - conf t
    - configure terminal
  - show run
    - displays all information from the running configuration
  - show run int
    - router port configuration
  - show run int tu#######
    - displays tunnel information requested by RHN
  - show run ne
    - router neighbours
  - show ip pin ne
    - neighbour check
  - show ip os ne
  - show ip os traf tu########
    - displays tunnel information
    - 0s = good; 1s = errors
  - show run int tu#######
  - show os ne
    - neighbour check
  - show run int tu#######
    - node information
    - useful for RHN verification
  - show os traffic tu########
    - ospf type(troubleshooting tx)
  - show run ospf
  - show run | s router ospf 21
    - ospf type
    - ***| = bar NOT the letter L
  - show run | s router ospfv3 21
    - ospf type(COLORLESS)
  - lway stop
  - lway stat
    - repeat until GUI displays modem activity
  - lway start
  - do wr
    - save configuration
  - (on modem)8+9+delete
    - zeroize modem

PMCS
Fault types
- deadline(X)

- inoperable/ unfit for field usage
- slash(/)
- minor faults

Fueling
- use fuel nozzle
- fuel in increments to avoid spills

HMMWV
- information plate(inside roadside door)
- oil spigot(2)
- tire pressure, wear, and expiration
- >5yrs = expired
- WWYY
- check for leaks
- ignition(wait for dash lights, then start)
- warmup(20min)
- check for leaks
- lights
- horn
- check trunk for ability to open/jams
- oil type: 15W-40

Trailer
- observe externals for faults
- front wheel
- tow mechanisms
- tent assembly

Generator
**pig tails configuration(to PDU)
- Good Whiskey BRO
- green - white - biack - red - orange
- ground - neutral - active - active - ?
- dead crank switch(s)
- 2kW contains 2 switches
- set to 'normal'
- ' emergency stop' out
- open front panel
- press CB1
- 'prime and run'
- listen for priming
- feel for rumble
- brief warmup period(30s)
- start
- listen for reduced engine rumble
- gauges
- check adjusters for normal activity
- voltage
- panel lights
- 'push test' lights
- check for leaks
- check for mechanical issues
- check tires
- turn off
- oil type: 15W-40
**2k oil type: 5W-30
**DO NOT USE CONVENIENCE RECEPTACLE

TRICONS(CONNEXs)
**1750s(1 per tricon)
**B I I (two set minimum per tricon)
- loaded last and free of egress
- triangles, fire extinguisher(s), shovels, sledge hammers, water cans

SINCGARS Radio

GK
- Half duplex: push to talk(release to listen)
- 1523 SINCGARS Radio
- Single Channel Ground and Airborne Radio System
- change mode
- pull knob and rotate
- Z: zeroizes
- ON: squelch on
- LD: changes frequencies/loading
- Test: self test
- STBY: low power; saving current state
- doesn't work without internal battery
- separate power source state overrides
- Single Channel(SC)
- standard
- Frequency Hop
- similar to FDMA
- SKL required
- timing sync

COMSEC
- Plain Text(PT)
- Cypher

Configuration(R/T-1523E)
- FCTN knob to Load(LD)
- 'menu' until 'mode'
- CHG to SC
- volume(9)
- CH1
- PT
- power: medium
- LD
- FREQ
- CLR
- '42000'
- STO